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Geometric Methods for Bayesian Inference and Modeling Discrete Data

Bernardo Williams

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Abstract

This thesis studies how geometry can be used to improve methods for approximate Bayesian inference and for modeling discrete data. In both cases the statistical object is non-Euclidean, but many practical methods assume a flat space structure. This thesis asks when geometry-aware modeling leads to more accurate, and in some cases, simpler methods. It also explores what computational tradeoffs such approaches might introduce.

The first part studies Bayesian inference, the task of approximating a posterior distribution. We consider Riemannian Laplace approximation and geometric Markov chain Monte Carlo methods for posterior distributions with strong curvature and multimodality. We show that geometry-aware methods improves the sample quality, but the metric must be chosen carefully. Metrics that better capture the geometry of the target space tend to be more accurate, while computationally efficient alternatives improve scalability at the cost of reduced accuracy.

The second part studies discrete data modeling, through inference of distributions on the probability simplex. We develop methods that rely on simplex-to-Euclidean bijections and stochastic interpolations. These bijections map the discrete data to continuous unbounded latent representations, while the interpolations allow for faithful recovery of the original categories. This construction allows the use of Euclidean modeling tools for

generation and classification. In particular we develop simplex-to-Euclidean flow matching for discrete data generation and a multiclass Gaussian Process classifier that has exact latent inference.

Throughout the thesis the main finding is that geometry-aware modeling can improve statistical performance. In Bayesian inference it gives accurate samples in curved and multimodal distributions, and in discrete data modeling, it simplifies the methodology by connecting to latent continuous representations.

Computing Reviews (2012) Categories and Subject Descriptors:

Mathematics of computing → Probability and statistics →
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Machine learning, Bayesian inference, generative models

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Laplace approximation, Markov chain Monte Carlo, flow models,
Gaussian process

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The last thing I will do is write the acknowledgments.

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List of Publications

This thesis is based on the following four original articles, referred throughout as Papers I-IV. The full articles are appended at the end of the thesis.

- I:** Hanlin Yu, Marcelo Hartmann, Bernardo Williams, Mark Girolami, and Arto Klami. Riemannian Laplace Approximation with the Fisher Metric. In *International Conference on Artificial Intelligence and Statistics*, pages 820–828. PMLR, 2024
- II:** Bernardo Williams, Hanlin Yu, Hoang Phuc Hau Luu, Georgios Arvanitidis, and Arto Klami. Geodesic Slice Sampler for Multimodal Distributions with Strong Curvature. In *Conference on Uncertainty in Artificial Intelligence*, pages 4543–4564. PMLR, 2025
- III:** Bernardo Williams, Victor M Yeom-Song, Marcelo Hartmann, and Arto Klami. Simplex-to-Euclidean Bijections for Categorical Flow Matching. In *International Conference on Artificial Intelligence and Statistics*, 2026. Accepted / to appear
- IV:** Bernardo Williams, Harsha Vardhan Tetali, Arto Klami, and Marcelo Hartmann. Simplex-to-Euclidean Bijection for Conjugate and Calibrated Multiclass Gaussian Process Classification, 2026. Under review

Author Contributions

Publication I: “Riemannian Laplace Approximation with the Fisher Metric” HY conceived the idea, wrote the code of the experiments and wrote the first draft of the manuscript. MH proposed the use of the Fisher-Rao metric in Riemann-Laplace method, and helped writing the theoretical derivations of central concepts in the paper. BW contributed to the code implementation. All authors jointly participated in the discussions and in the writing of the manuscript.

Publication II: “Geodesic Slice Sampler for Multimodal Distributions with Strong Curvature” BW conceived the idea, wrote the code of the experiments, ran all the experiments, and wrote the first draft of the manuscript. The theoretical analysis was initially done by BW and corrected and improved with help from HY, HPHL and MH. The writing of the paper was done as joint collaboration of all authors, with AK and MH providing supervision and guidance through the whole process. MH additionally introduced external collaboration for BW, which benefited paper writing and development: GA suggested the teaser image and changes to text.

Publication III: “Simplex-to-Euclidean bijections for Categorical Flow Matching” BW conceived the paper’s idea, wrote the code of the experiments and wrote the first draft of the manuscript. BW executed all of the experiments and derived the theoretical results. VYS implemented the comparison methods (SEDD) and measured running times of the methods (App. C.3). Jointly all authors participated in the discussion and writing of the paper.

Publication IV: “Simplex-to-Euclidean Bijection for Conjugate and Calibrated Multiclass Gaussian Process Classification” BW and MH jointly conceived the paper’s original idea. BW wrote the code of the experiments, wrote the first draft of the manuscript, and derived the theoretical results. HT and MH proof read the propositions and theorems. All of the authors participated in the discussion and writing.

Chapter 1

Introduction

This thesis studies geometric methods for two core tasks in machine learning: Bayesian inference and modeling of discrete data. In both cases we start with a probabilistic model whose unknown parameters are to be estimated in a principled way. In Bayesian inference [Gelman et al., 1995], knowledge is given as a prior distribution, which is updated with the observed data, resulting in a posterior distribution through Bayes’ rule. In a few special cases (such as classic Gaussian process regression) the posterior distribution is analytically tractable, that is, the model is conjugate, and inference of the parameters reduces to drawing samples from this distribution. Most models, however, the distribution is intractable and we rely on approximate inference techniques to estimate the parameters.

Recently, machine learning places emphasis on generative modeling, where the objective is to use the available data to estimate the unknown parameters and then draw new samples from the fitted model. The process is applied to generate images, videos, molecules or text for example [Lipman et al., 2024]. A common technique is modeling the transport map from a simple base distribution (e.g. Gaussian) to the data distribution [Lipman et al., 2023, Albergo and Vanden-Eijnden, 2023, Liu et al., 2023]. The task of generation becomes drawing samples from the simple base and transporting them to the approximate data distribution with the inferred model. This strategy is naturally defined for continuous data, but it can also be applied to discrete data through continuous relaxations, provided we have a reliable way to map generated points back to discrete outcomes.

This thesis studies a common theme: geometry [Lee, 2018, Amari and Nagaoka, 2000]. The algorithms for both tasks need to operate in some geometry to measure distances, and to follow trajectories that adhere to the shape statistical object. Often there is no attention to the geometry, instead Euclidean geometry is assumed by convention. Posterior distributions can

have strong local curvature [Girolami and Calderhead, 2011] or multiple separated modes [Lan et al., 2014]. Discrete probability vectors lie on the simplex, which has its own intrinsic geometry [Aitchison, 1986]. Ignoring this structure often leads to methods that are simple but inaccurate.

Riemannian geometry [Do Carmo and Flaherty Francis, 1992, Lee, 2018] provides tools to encode local structure through a metric and to build algorithms that adhere to it. The metric choice introduces a tradeoff: richer geometry can improve statistical behavior while often increasing the computational cost, since even a simple operation like measuring the distance between two points may now require integration. For practical methods, geometry must therefore be chosen not only for faithfulness to the target distribution, but also for computational efficiency and tractability.

In Bayesian inference, let us consider the problem of sampling from curved unimodal posterior distributions. Girolami and Calderhead [2011] proposed adapting to the posterior geometry by using the Fisher information metric, which captures local structure from the Kullback-Leibler divergence. The problem is that the metric is expensive to compute, it requires problem specific, and it is intractable for some distributions. Alternatively, metrics based on the geometry of the posterior graph [Hartmann et al., 2022, 2023] can be computationally cheaper but are not completely faithful to the underlying geometry. Paper I studies the balance between these two metric choices. An adjacent problem is multimodal sampling. While geometry-aware samplers can improve exploration of strongly curved targets, their application to multimodal targets is less developed than tempering and diffusion based variants [Lan et al., 2014, Łatuszyński et al., 2025]. Paper II offers a new sampling strategy and metric choice for curved and multimodal targets. In Bayesian inference, the choice of metric is a modeling decision (curved and/or multimodal) and a computational one.

In modeling discrete data, geometry naturally arises [Fisher, 1925, Aitchison, 1986]. The simplex is a compact convex subset of $\mathbb{R}^K$ defined by non-negative components that sum up to one. This space can be given the Fisher information geometry of the categorical distribution, but this approach implicitly models distributions on a compact space. Since much work has been done on Euclidean space, a bijection that respects the geometry of the simplex and connects to Euclidean space eases modeling. The log-ratio geometry between the elements is known as Aitchison geometry and provides simplex-to-Euclidean bijections. These maps allow for generative modeling and classification in the simplex via a latent Euclidean representation. We develop stochastic interpolations that convert discrete data to open simplex distributions, since the bijections are defined on the

interior. In Bayesian classification the posterior distribution is usually intractable, but the the bijections with the stochastic interpolation makes it possible to fit a conjugate model in the latent space. Papers III and IV focus on generative modeling and classification respectively.

Research questions

The following questions summarize the high level themes of the dissertation:

1. How can the geometric structure of the target density be used to improve approximate Bayesian inference?
2. How can geometry-aware sampling methods better handle targets with both strong curvature and multimodality?
3. How can the simplex geometry be connected to Euclidean representations for modeling discrete data?
4. Can the Aitchison geometry enable simpler and exact inference procedures in multiclass-classification?

The aim of the thesis is to advance geometric methods both in the context of approximate Bayesian inference and modeling discrete data, by developing methods that use geometry in novel ways. Concretely, the Bayesian inference part studies geometry-aware posterior approximation and sampling through Riemannian Laplace approximation (Paper I) and geodesic slice sampling (Paper II). The discrete-modeling part studies simplex-to-Euclidean bijections and stochastic interpolations for discrete generation (Paper III) and conjugate multiclass Gaussian process classification (Paper IV).

The dissertation is organized as follows. Chapter 2 provides the background on Riemannian geometry, approximate Bayesian inference and modeling discrete data that our contributions build on. The core content is then divided into two main blocks. The first block (Chapter 3) focuses on Bayesian inference, studying geometric improvements for posterior approximation and sampling. The second block (Chapter 4) focuses on modeling discrete data, using the simplex geometry to build methods that connect naturally to Euclidean tools. Across both parts, the main objective is the same: efficient geometric modeling that improves statistical performance.

Summary of the contributions

This thesis makes the following methodological contributions:

- The thesis proposes modifying the choice of the Riemannian metric tensor in two Bayesian inference methods: Riemannian Laplace approximation (Paper I) and geodesic slice sampler (Paper II).
- The Riemannian Laplace approximation with the Monge metric has an inherent bias on Gaussian targets, Paper I proposes two techniques to correct the bias: (i) using the logarithm map, and (ii) using the Fisher information as the metric tensor.
- Paper II adapts the geodesic slice sampler, which was previously used for manifolds with closed form geodesics, to target distributions defined on Euclidean space. It also proposes new Riemannian metrics constructed from the target density for multimodal sampling.
- The thesis proposes using simplex-to-Euclidean bijections and stochastic interpolations for modeling discrete data in Euclidean space instead of the simplex. Two different problems are studied: discrete data generation (Paper III) and Gaussian process classification (Paper IV).
- Based on the Euclidean representations, the thesis proposes using general purpose latent Euclidean models for discrete data, including flow-based generative models (Paper III) and Gaussian process regression models for classification (Paper IV).

Notation

This page collects the most important notation organized by topic.

Geometric Bayesian inference

$\mathbf{x} \in \mathbb{R}^D, p(\mathbf{x}),$	Variable of interest and target density.
$\ell(\mathbf{x}) := \log p(\mathbf{x})$	Log-target density.
$\hat{\mathbf{x}}$	Mode or MAP estimate of the target.
$\mathbf{G}(\mathbf{x})$	Riemannian metric tensor.
$\text{Exp}_{\mathbf{x}}, \text{Log}_{\mathbf{x}}$	Exponential and logarithm maps induced by the metric.
$\gamma_{\mathbf{x}, \mathbf{v}}(t)$	Geodesic at time t with initial conditions $\mathbf{x}, \mathbf{v}$
$p_{\mathcal{H}}(\mathbf{x})$	Hausdorff density associated with the Riemannian volume.
$\Gamma_{ij}^k(\mathbf{x})$	Christoffel symbols of the second kind.

Gaussian Processes

$\mathbf{f}(\cdot) \sim \mathcal{GP}(\mathbf{0}, K_{\theta})$	Latent Gaussian process prior with zero mean and kernel K_{θ} .
$\boldsymbol{\mu}_*, \boldsymbol{\Sigma}_*$	predictive mean, and predictive covariance in the GP model for point $\mathbf{x}_*$.
σ^2	Latent Gaussian noise variance.

Latent constructions

$K, D := K - 1$	Number of categories and corresponding latent dimension.
$c_i, \mathbf{x}_i, \mathbf{z}_i, \mathbf{r}_i$	Class label $c_i \in \{1, \dots, K\}$, point in open simplex, point in Euclidean space and regressor.
$\Delta^D, \mathring{\Delta}^D$	Closed simplex and open simplex.
φ, φ^{-1}	Bijection between the open simplex and Euclidean latent space.
$\langle \cdot, \cdot \rangle_A, d_A(\cdot, \cdot)$	Aitchison inner product and Aitchison distance on the simplex.
$\mathcal{R}_k$	Strict arg max region associated with category k .
$\pi, \hat{\pi}$	True and estimated categorical probabilities.
λ, α	Interpolation weight and Dirichlet noise parameter.
$\boldsymbol{\mu}^{(k)}, \mathbf{m}^{(k)} = \varphi(\boldsymbol{\mu}^{(k)})$	Simplex and latent Euclidean representatives of class k .
p_0, p_1, p_t	Base distribution, data distribution, and marginal distribution at time t .
$\mathbf{u}_t(\cdot \mathbf{z}), \mathbf{v}_t^\theta(\cdot)$	Target conditional velocity field and parametric velocity field.

Chapter 2

Background

This chapter introduces the background needed for the methods developed in Chapters 3 and 4. We start by giving a brief introduction to Riemannian geometry, then introducing methods for approximate Bayesian inference and finally approaches for modeling discrete data.

2.1 Basic notions in Riemannian Geometry

For a formal and rigorous introduction to Riemannian geometry we refer the reader to Do Carmo and Flaherty Francis [1992], Lee [2018]. Here we introduce the only the concepts needed later in the thesis, under simplified settings that cover our examples. We consider manifolds that admit a global chart, or are smooth embedded submanifolds of Euclidean space. This allows us to write the coordinates globally, and avoid dealing with local coordinates patches. When the space has a boundary (e.g. the simplex) we consider the differential geometric objects only on the interior of the space.

2.1.1 Riemannian manifolds

Manifolds and tangent spaces A smooth D -dimensional manifold $\mathcal{M}$ is a topological space that is locally diffeomorphic to $\mathbb{R}^D$. For simplicity, we assume that $\mathcal{M}$ admits a global chart $\varphi : \mathcal{M} \rightarrow U \subseteq \mathbb{R}^D$ with inverse $\psi = \varphi^{-1}$. For $\mathbf{x} \in \mathcal{M}$, we denote its coordinates by $\mathbf{z} = \varphi(\mathbf{x}) \in U$, where $z^j : \mathcal{M} \rightarrow \mathbb{R}$ denotes the j -th coordinate function, i.e. $\varphi(\mathbf{x}) = (z^1(\mathbf{x}), \dots, z^D(\mathbf{x}))$. The tangent space $T_{\mathbf{x}}\mathcal{M}$ is the vector space of velocities of smooth curves that pass through $\mathbf{x}$. If $\gamma : (-\varepsilon, \varepsilon) \rightarrow \mathcal{M}$ is a smooth curve with $\gamma(0) = \mathbf{x}$, let $\mathbf{z}(t) = \varphi(\gamma(t))$. Then $\dot{\gamma}(t) = \psi(\dot{\mathbf{z}}(t))$, and by the chain

rule

$$\dot{\gamma}(0) = \sum_{i=1}^D \dot{z}^i(0) \frac{\partial}{\partial z^i},$$

where $\frac{\partial}{\partial z^i} := \frac{\partial \psi}{\partial z^i}(\mathbf{z}) \in T_{\mathbf{x}}\mathcal{M}$. The expression above shows that $\partial/\partial z^i$ span the tangent space. Moreover,

$$\frac{\partial}{\partial z^i}(z^j) = \frac{\partial(z^j \circ \psi)}{\partial z^i}(\mathbf{z}) = \frac{\partial z^j}{\partial z^i} = \delta_i^j,$$

which implies linear independence, so they form a basis of $T_{\mathbf{x}}\mathcal{M}$.

Riemannian metric If two tangent vectors $\mathbf{u}, \mathbf{v} \in T_{\mathbf{x}}\mathcal{M}$, we write $\mathbf{u} = \sum_{i=1}^D u^i \frac{\partial}{\partial z^i}, \mathbf{v} = \sum_{i=1}^D v^i \frac{\partial}{\partial z^i}$. The Riemannian metric defines for each $\mathbf{x} \in \mathcal{M}$ an inner product on the tangent space $g : T_{\mathbf{x}}\mathcal{M} \times T_{\mathbf{x}}\mathcal{M} \rightarrow \mathbb{R}$. We represent it with a symmetric positive definite matrix $\mathbf{G}(\mathbf{x})$ with $G_{ij}(\mathbf{x}) = g_{\mathbf{x}}(\partial/\partial z^i, \partial/\partial z^j)$, such that

$$g_{\mathbf{x}}(\mathbf{u}, \mathbf{v}) = \langle \mathbf{u}, \mathbf{v} \rangle_{\mathbf{G}(\mathbf{x})} = \mathbf{u}^\top \mathbf{G}(\mathbf{x}) \mathbf{v},$$

where on the right-hand side $\mathbf{u} = (u^1, \dots, u^D)^\top$ and $\mathbf{v} = (v^1, \dots, v^D)^\top$ denote the vector components. This metric defines lengths of tangent vectors, lengths of curves, angles, and geodesic distance. A manifold endowed with a metric tensor is called a *Riemannian manifold* and denoted by $(\mathcal{M}, g)$.

Embedded manifolds and the pullback metric Some of the manifolds we consider are smooth embedded submanifolds of $\mathbb{R}^K$. Let $\psi : U \subseteq \mathbb{R}^D \rightarrow \mathbb{R}^K$ be a smooth embedding and let $\mathcal{M} = \psi(U)$. Then, for $\mathbf{x} = \psi(\mathbf{z}) \in \mathcal{M}$, the tangent space is

$$T_{\mathbf{x}}\mathcal{M} = \{ \mathbf{J}_\psi(\mathbf{z}) \mathbf{v} : \mathbf{v} \in \mathbb{R}^D \},$$

where $\mathbf{J}_\psi$ is the Jacobian matrix of ψ . The ambient inner product induces a Riemannian metric on $\mathcal{M}$ by restriction to the tangent space. If $\mathbf{u}, \mathbf{v} \in T_{\mathbf{x}}\mathcal{M}$, such that $\mathbf{u} = \mathbf{J}_\psi(\mathbf{z}) \mathbf{a}$ and $\mathbf{v} = \mathbf{J}_\psi(\mathbf{z}) \mathbf{b}$ for $\mathbf{a}, \mathbf{b} \in \mathbb{R}^D$, then $g_{\mathbf{x}}(\mathbf{u}, \mathbf{v}) = \langle \mathbf{u}, \mathbf{v} \rangle_2 = \mathbf{a}^\top \mathbf{J}_\psi(\mathbf{z})^\top \mathbf{J}_\psi(\mathbf{z}) \mathbf{b}$. In coordinates $\mathbf{z} \in U$, this induced metric is represented by the matrix

$$\mathbf{G}(\mathbf{z}) = \mathbf{J}_\psi(\mathbf{z})^\top \mathbf{J}_\psi(\mathbf{z}).$$

Let us consider a classic example. The graph of a smooth function $f : \mathbb{R}^D \rightarrow \mathbb{R}$ is:

$$\mathcal{G}_f := \{(\mathbf{z}, f(\mathbf{z})) : \mathbf{z} \in \mathbb{R}^D\} \subset \mathbb{R}^{D+1},$$

Name	Definition	Embedded	Metric
Sphere	$\mathbb{S}_+^D := \{\mathbf{x} \in \mathbb{R}_{\geq 0}^K : \sum_{k=1}^K x_k^2 = 1\}$	✓	Pullback
Simplex	$\Delta^D := \{\mathbf{x} \in \mathbb{R}_{\geq 0}^K : \sum_{k=1}^K x_k = 1\}$	✓	Custom
Graph	$\mathcal{G}_\alpha := \{(\mathbf{x}, \alpha \ell(\mathbf{x})) : \mathbf{x} \in \mathbb{R}^D\}$	✓	Pullback
Statistical	$\Theta := \{\mathbf{x} \in \mathbb{R}^D : p_{\mathcal{Y} \mathcal{X}}(\mathbf{y} \mathbf{x}) p_{\mathcal{X}}(\mathbf{x}) > 0\}$	–	Custom

Table 2.1: Summary of the manifolds considered in the thesis. In the case of the simplex and statistical manifolds we equip the space with custom choices metric tensors presented in Chapter 3 and Chapter 4.

the embedding is $\psi(\mathbf{z}) = (\mathbf{z}, f(\mathbf{z}))$, so the induced metric is

$$\mathbf{G}(\mathbf{z}) = \mathbf{I}_D + \nabla f(\mathbf{z}) \nabla f(\mathbf{z})^\top.$$

If $f(\mathbf{z}) = \alpha \ell(\mathbf{z})$ for a log density ℓ and tempering parameter $\alpha \geq 0$, this is the Monge metric (see Chapter 3). The different manifolds considered in this thesis are summarized in Table 2.1

2.1.2 Geodesics

The metric tensor defines the length of a smooth curve $\gamma : [0, 1] \rightarrow \mathcal{M}$ by

$$L(\gamma) = \int_0^1 \|\dot{\gamma}(t)\|_{\mathbf{G}(\gamma(t))} dt. \quad (2.1)$$

A geodesic between two points $\mathbf{x}_0, \mathbf{x}_1 \in \mathcal{M}$ is the curve that has the shortest length, namely

$$d_g(\mathbf{x}_0, \mathbf{x}_1) = \inf_{\gamma} \int_0^1 \|\dot{\gamma}(t)\|_{\mathbf{G}(\gamma(t))} dt, \quad \gamma(0) = \mathbf{x}_0, \gamma(1) = \mathbf{x}_1. \quad (2.2)$$

A geodesic can equivalently be characterized as the curve with zero covariant acceleration, for this we need to define the notion of covariant derivative.

Levi-Civita connection The Levi-Civita connection provides an intrinsic derivative of vector fields along tangent directions. It is the unique affine connection that is torsion free and compatible with the metric [Lee, 2018]. We use the notation $\partial_i := \partial/\partial z^i$. If $\mathbf{V} = \sum_{j=1}^D V^j \partial_j$ is a vector field and $\mathbf{u} = \sum_{i=1}^D u^i \partial_i \in T_{\mathbf{x}}\mathcal{M}$, then

$$\nabla_{\mathbf{u}}^{\mathbf{G}} \mathbf{V} = \sum_{k=1}^D \left(\sum_{i=1}^D u^i \partial_i V^k + \sum_{i,j=1}^D \Gamma_{ij}^k u^i V^j \right) \partial_k,$$

where the Christoffel symbols of the second kind

$$\Gamma_{ij}^k(\mathbf{x}) = \frac{1}{2} \sum_{m=1}^D \mathbf{G}_{km}^{-1}(\mathbf{x}) (\partial_i \mathbf{G}_{mj}(\mathbf{x}) + \partial_j \mathbf{G}_{im}(\mathbf{x}) - \partial_m \mathbf{G}_{ij}(\mathbf{x})). \quad (2.3)$$

Exponential map A geodesic γ has vanishing covariant acceleration, $\nabla_{\dot{\gamma}(t)}^{\mathbf{G}} \dot{\gamma}(t) = 0$, this condition gives the following differential equation:

$$\ddot{\gamma}^k(t) + \sum_{i,j=1}^D \Gamma_{ij}^k(\gamma(t)) \dot{\gamma}^i(t) \dot{\gamma}^j(t) = 0, \quad k = 1, \dots, D, \quad (2.4)$$

where we used $\left(\sum_{i=1}^D \dot{\gamma}^i(t) \partial_i \right) \dot{\gamma}^k(t) = \frac{d}{dt} \dot{\gamma}^k(t) = \ddot{\gamma}^k(t)$.

For $\mathbf{x} \in \mathcal{M}$ and $\mathbf{v} \in T_{\mathbf{x}}\mathcal{M}$, let $\gamma_{\mathbf{x},\mathbf{v}}$ be geodesic with initial conditions $\gamma_{\mathbf{x},\mathbf{v}}(0) = \mathbf{x} \in \mathcal{M}$ and $\dot{\gamma}_{\mathbf{x},\mathbf{v}} = \mathbf{v} \in T_{\mathbf{x}}\mathcal{M}$. The exponential map is defined as

$$\text{Exp}_{\mathbf{x}}(t\mathbf{v}) = \gamma_{\mathbf{x},\mathbf{v}}(t).$$

If the system does not have analytical solution, the exponential map is computed with numerical integration using standard ordinary differential equation solvers, such as Euler, or Dormand-Prince's method [Kidger, 2022].

Logarithm map The logarithm map is the inverse of the exponential map, given two close enough positions $\mathbf{x}_0, \mathbf{x}_1 \in \mathcal{M}$ that the inverse is well defined, then velocity that generates the geodesic connecting them. This a boundary value problem whose solution gives

$$\mathbf{v} = \text{Log}_{\mathbf{x}_0}(\mathbf{x}_1) \in T_{\mathbf{x}_0}\mathcal{M},$$

such that $\text{Exp}_{\mathbf{x}_0}(\mathbf{v}) = \mathbf{x}_1$. If the solution is not computable analytically, the problem can be solved numerically by standard boundary value solvers, such as Virtanen et al. [2020].

The main computational bottleneck comes from the computation of the Christoffel symbols since they involve the inverse metric and derivatives of the metric. The choice of numerical solver affects accuracy and runtime: fixed or adaptive step-size, and low to higher order solvers introduce tradeoff between stability, numerical error and computational cost.

2.1.3 Riemannian gradient and Hessian

The notions of gradients, Hessian are modified such that they are consistent with the geometry introduced by $\mathbf{G}(\mathbf{x})$, and are useful later in Chapter 3.

Riemannian gradient Let $f : \mathcal{M} \rightarrow \mathbb{R}$ be a smooth function. The Riemannian gradient $\nabla_{\mathbf{G}} f(\mathbf{x}) \in T_{\mathbf{x}}\mathcal{M}$ is tangent vector which satisfies the identity

$$\langle \nabla_{\mathbf{G}} f(\mathbf{x}), \mathbf{v} \rangle_{\mathbf{G}(\mathbf{x})} = \mathrm{d}f_{\mathbf{x}}(\mathbf{v}), \quad \forall \mathbf{v} \in T_{\mathbf{x}}\mathbb{R}^D,$$

where the differential $\mathrm{d}f_{\mathbf{x}} : T_{\mathbf{x}}\mathbb{R}^D \rightarrow \mathbb{R}$ such that $\mathrm{d}f_{\mathbf{x}}(\mathbf{v}) = \sum_{i=1}^D \frac{\partial f}{\partial z^i}(\mathbf{x}) v^i$. Solving for $\nabla_{\mathbf{G}}$, the Riemannian gradient has the components

$$(\nabla_{\mathbf{G}} f)^i = \sum_{j=1}^D G^{ij} \partial_j f, \quad (2.5)$$

where G^{ij} are the components of $\mathbf{G}^{-1}(\mathbf{x})$. If f has support on $\mathcal{M} \subset \mathbb{R}^D$, the components are the Euclidean gradient scaled by the inverse metric. This quantity is sometimes referred as the natural gradient [Amari, 1998], in this case we can collect the components into the vector

$$\mathbf{G}(\mathbf{x})^{-1} \nabla_{\mathbf{x}} f,$$

where $\nabla_{\mathbf{x}} f$ is the Euclidean gradient.

Riemannian Hessian The Riemannian Hessian is defined as the covariant derivative of the Riemannian gradient, for $\mathbf{x} \in \mathcal{M}$ and $\mathbf{v} \in T_{\mathbf{x}}\mathcal{M}$

$$\text{Hess}_{\mathbf{G}} f(\mathbf{x})[\mathbf{v}] := \nabla_{\mathbf{v}}^{\mathbf{G}} \nabla_{\mathbf{G}} f.$$

This expression can be viewed as the vector with components:

$$(\text{Hess}_{\mathbf{G}} f(\mathbf{x})[\mathbf{v}])^i = \sum_{j=1}^D \sum_{m=1}^D G^{im} \left(\partial_m \partial_j f - \sum_{k=1}^D \Gamma_{mj}^k \partial_k f \right) v^j.$$

If f has support on $\mathcal{M} \subset \mathbb{R}^D$, we can collect these components in the matrix

$$\mathbf{G}^{-1}(\mathbf{x}) \left(\nabla^2 f(\mathbf{x}) - \sum_{k=1}^D \mathbf{\Gamma}^k(\mathbf{x}) \partial_k f(\mathbf{x}) \right) \mathbf{v}, \quad (2.6)$$

where $\nabla^2 f(\mathbf{x})$ is the Euclidean Hessian and $[\mathbf{\Gamma}^k(\mathbf{x})]_{ij} = \Gamma_{ij}^k(\mathbf{x})$. For Euclidean metric the Riemannian Hessian is a Hessian vector product.

2.2 Approximate Bayesian inference techniques

Let $\mathcal{X}$ be a continuous random variable on $\mathbb{R}^D$ with smooth density $p_{\mathcal{X}}(\mathbf{x})$ with respect to the Lebesgue measure. Let $\mathbf{y}$ denote observed data, with likelihood $p_{\mathcal{Y}|\mathcal{X}}(\mathbf{y} | \mathbf{x})$. In Bayesian inference, the posterior density is

$$\underbrace{p_{\mathcal{X}|\mathcal{Y}}(\mathbf{x} | \mathbf{y})}_{\text{posterior}} = \frac{\overbrace{p_{\mathcal{Y}|\mathcal{X}}(\mathbf{y} | \mathbf{x})}^{\text{likelihood}} \overbrace{p_{\mathcal{X}}(\mathbf{x})}^{\text{prior}}}{\underbrace{p_{\mathcal{Y}}(\mathbf{y})}_{\text{evidence}}}, \quad (2.7)$$

and the marginal likelihood or model evidence is

$$p_{\mathcal{Y}}(\mathbf{y}) = \int_{\mathbb{R}^D} p_{\mathcal{Y}|\mathcal{X}}(\mathbf{y} | \mathbf{x}) p_{\mathcal{X}}(\mathbf{x}) d\mathbf{x}.$$

When the posterior falls on the same parametric family as the prior, the model is called conjugate and we do not need to rely on approximations. For example, we can compute expectations and credible intervals directly from this distribution. In most cases, however, $p_{\mathcal{Y}}(\mathbf{y})$ is intractable and we therefore work with the unnormalized target density

$$p(\mathbf{x}) := p_{\mathcal{Y}|\mathcal{X}}(\mathbf{y} | \mathbf{x}) p_{\mathcal{X}}(\mathbf{x}). \quad (2.8)$$

Approximate Bayesian inference then relies on $p(\mathbf{x})$ or on its log-density $\ell(\mathbf{x}) := \log p(\mathbf{x})$ to form approximations of the posterior (2.7).

In this thesis, the practical objective is to draw samples from the posterior distribution. Since the target density is proportional to the posterior, sampling from either yields the same distribution. Given a set of samples we can estimate posterior quantities by Monte Carlo, for example, if $\{\mathbf{x}^{(s)}\}_{s=1}^S$ are samples distributed according to the target distribution, then for a measurable function h we can estimate expectations by

$$\mathbb{E}[h(\mathcal{X})] \approx \frac{1}{S} \sum_{s=1}^S h(\mathbf{x}^{(s)}). \quad (2.9)$$

This thesis focuses on two major families of approximate inference methods. The first consists of **distributional approximation methods**, where we approximate the posterior with a tractable approximation. In this family, we discuss variational inference and Laplace approximation. The second is **Markov chain Monte Carlo methods**, where we construct a Markov chain whose stationary distribution is the posterior. We discuss several such methods in Section 2.4. The following sections introduce these two approximate Bayesian inference approaches.

2.3 Distributional approximations

Distributional approximations replace $p_{\mathcal{X}|\mathcal{Y}}(\mathbf{x} \mid \mathbf{y})$ with a tractable distributional family $q_{\boldsymbol{\theta}}(\mathbf{x})$.

Laplace approximation The idea behind Laplace approximation is to approximate the log-target distribution around its mode [MacKay, 2003]. Assume that the target density has its mode at value $\hat{\mathbf{x}}$ and take $\ell(\mathbf{x}) = \log p(\mathbf{x})$, then a second order Taylor expansion of $\ell(\mathbf{x})$ at the mode gives

$$\ell(\mathbf{x}) = \ell(\hat{\mathbf{x}}) + \langle \nabla \ell(\hat{\mathbf{x}}), \mathbf{x} - \hat{\mathbf{x}} \rangle + \frac{1}{2} \|\mathbf{x} - \hat{\mathbf{x}}\|_{\nabla^2 \ell(\hat{\mathbf{x}})}^2 + \mathcal{O}(\|\mathbf{x} - \hat{\mathbf{x}}\|^3). \quad (2.10)$$

At the mode $\nabla \ell(\hat{\mathbf{x}}) = 0$ and the inner product term vanishes, therefore the posterior is approximated by an unnormalized Gaussian

$$p(\mathbf{x}) \approx p(\hat{\mathbf{x}}) \exp\left(-\frac{1}{2} \|\mathbf{x} - \hat{\mathbf{x}}\|_{\boldsymbol{\Sigma}^{-1}}^2\right), \quad (2.11)$$

where $\boldsymbol{\Sigma}^{-1} = -\nabla^2 \ell(\hat{\mathbf{x}})$ and the notation means $\|\mathbf{x}\|_{\boldsymbol{\Sigma}^{-1}}^2 = \mathbf{x}^\top \boldsymbol{\Sigma}^{-1} \mathbf{x}$. This is the kernel of the Gaussian distribution $\mathcal{N}(\hat{\mathbf{x}}, \boldsymbol{\Sigma})$. Approximate samples are drawn from $q(\mathbf{x})$. This approximation is exact for Gaussian distributions. If $\mathcal{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then the mode of the density $p_{\mathcal{X}}(\mathbf{x})$ is $\boldsymbol{\mu}$ and the Hessian of the log density is $\nabla^2 \log p_{\mathcal{X}}(\mathbf{x}) = -\boldsymbol{\Sigma}^{-1}$, which coincides with Laplace approximation.

Variational inference Variational inference approximates the target $p(\mathbf{x})$ with a parametric family $q_{\boldsymbol{\theta}}(\mathbf{x})$ according to a measure of divergence between the distributions [Blei et al., 2017]. Let us consider a common choice, Kullback-Leibler (KL) divergence, defined as

$$\text{KL}(p(\mathbf{x}) \| q(\mathbf{x})) := \mathbb{E}_{\mathbf{x} \sim p} \left[\log \frac{p(\mathbf{x})}{q(\mathbf{x})} \right],$$

which is not symmetric in its arguments, and q must be non-zero on the support of p . When only the target is available we can resort to reverse KL

$$\begin{aligned} \text{KL}(q_{\boldsymbol{\theta}}(\mathbf{x}) \| p(\mathbf{x})) &= \mathbb{E}_{\mathbf{x} \sim q_{\boldsymbol{\theta}}} [\log q_{\boldsymbol{\theta}}(\mathbf{x}) - \log p(\mathbf{x})] \\ &\approx \frac{1}{N} \sum_{i=1}^N \left[\log q_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}) - \log p(\mathbf{x}^{(i)}) \right]. \end{aligned}$$

The reverse KL objective is approximated by Monte Carlo samples drawn as $\mathbf{x}^{(i)} \sim q_{\boldsymbol{\theta}}(\mathbf{x})$. The parameters $\boldsymbol{\theta}$ are optimized to minimize this objective.

The second term $\log p(\mathbf{x}^{(i)})$ does not vanish under derivative computation with respect to $\boldsymbol{\theta}$ since the generation of samples $\mathbf{x}^{(i)}$ depends on the parameters $\boldsymbol{\theta}$. A common method for reducing variance is reparameterization, if the distributional family $q_{\boldsymbol{\theta}}(\mathbf{x}^{(i)})$ allows for $\mathbf{x} = \varphi_{\boldsymbol{\theta}}(\mathbf{z})$ with $\mathbf{z} \sim q_{\mathbf{z}}$ independent of $\boldsymbol{\theta}$, the objective can be rewritten as

$$\text{KL}(q_{\boldsymbol{\theta}}(\mathbf{x})\|p(\mathbf{x})) = \mathbb{E}_{\mathbf{z} \sim q_{\mathbf{z}}} [\log q_{\boldsymbol{\theta}}(\varphi_{\boldsymbol{\theta}}(\mathbf{z})) - \log p(\varphi_{\boldsymbol{\theta}}(\mathbf{z}))]. \quad (2.12)$$

A choice for the approximation is the mean field variational family which reduces the number of parameters by assuming decomposition over the dimensions

$$q_{\boldsymbol{\theta}}(\mathbf{x}) = \prod_{j=1}^D q_{\boldsymbol{\theta}_j}(x_j),$$

such that $\boldsymbol{\theta} = (\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_D)$. This reduces the number of parameters but loses modeling flexibility.

Let us again consider a Bayesian model where the posterior density is $p_{\mathcal{X}|\mathcal{Y}}(\mathbf{x} | \mathbf{y})$ (Eq. 2.7). In this case minimizing the KL divergence between $q_{\boldsymbol{\theta}}$ and $p_{\mathcal{X}|\mathcal{Y}}(\mathbf{x} | \mathbf{y})$ is equivalent to maximizing a lower bound of the marginal log likelihood (evidence),

$$\begin{aligned} \text{KL}(q_{\boldsymbol{\theta}}(\mathbf{x})\|p_{\mathcal{X}|\mathcal{Y}}(\mathbf{x} | \mathbf{y})) &= \mathbb{E}_{\mathbf{x} \sim q_{\boldsymbol{\theta}}} [\log q_{\boldsymbol{\theta}}(\mathbf{x}) - \log p_{\mathcal{X}|\mathcal{Y}}(\mathbf{x} | \mathbf{y})] \\ &= \mathbb{E}_{q_{\boldsymbol{\theta}}} [\log q_{\boldsymbol{\theta}}(\mathbf{x}) - \log p_{\mathcal{Y}|\mathcal{X}}(\mathbf{y} | \mathbf{x}) - \log p_{\mathcal{X}}(\mathbf{x}) + \log p_{\mathcal{Y}}(\mathbf{y})] \\ &= \log p_{\mathcal{Y}}(\mathbf{y}) - (\mathbb{E}_{\mathbf{x} \sim q_{\boldsymbol{\theta}}} [\log p_{\mathcal{Y}|\mathcal{X}}(\mathbf{y} | \mathbf{x})] - \text{KL}(q_{\boldsymbol{\theta}}(\mathbf{x})\|p_{\mathcal{X}}(\mathbf{x}))). \end{aligned}$$

Since $\text{KL}(q_{\boldsymbol{\theta}}(\mathbf{x})\|p_{\mathcal{X}|\mathcal{Y}}(\mathbf{x} | \mathbf{y})) \geq 0$ the following inequality holds:

$$\log p_{\mathcal{Y}}(\mathbf{y}) \geq \mathbb{E}_{\mathbf{x} \sim q_{\boldsymbol{\theta}}} [\log p_{\mathcal{Y}|\mathcal{X}}(\mathbf{y} | \mathbf{x})] - \text{KL}(q_{\boldsymbol{\theta}}(\mathbf{x})\|p_{\mathcal{X}}(\mathbf{x})),$$

The right hand side is called the evidence lower bound (ELBO), and minimizing the KL is equivalent to maximizing the ELBO

$$\mathcal{L}_{\text{ELBO}}(\boldsymbol{\theta}) := \mathbb{E}_{\mathbf{x} \sim q_{\boldsymbol{\theta}}} [\log p_{\mathcal{Y}|\mathcal{X}}(\mathbf{y} | \mathbf{x})] - \text{KL}(q_{\boldsymbol{\theta}}(\mathbf{x})\|p_{\mathcal{X}}(\mathbf{x})). \quad (2.13)$$

The ELBO objective is used to optimize the variational parameters. In Section 2.5.2 we use the same approach to derive lower bounds of the marginal log likelihood for sparse Gaussian processes.

2.4 Markov chain Monte Carlo

Markov chain Monte Carlo (MCMC) methods construct a chain of dependent samples that approximate the target distribution [Robert et al., 2004].

Let $\pi(\mathbf{x})$ denote the normalized target density, with $\pi(\mathbf{x}) := p(\mathbf{x}) / \int p(\mathbf{x}) d\mathbf{x}$, and let $T(\mathbf{x}, \mathbf{x}')$ denote a transition kernel, the probability of transition from $\mathbf{x}$ to $\mathbf{x}'$. A Markov chain is specified by an initial distribution $p^{(0)}(\mathbf{x})$ and evolves as

$$\pi^{(t+1)}(\mathbf{x}') = \int_{\mathbb{R}^D} \pi^{(t)}(\mathbf{x}) T(\mathbf{x}, \mathbf{x}') d\mathbf{x}.$$

A Markov chain is constructed to satisfy two properties [MacKay, 2003]:

1. **Stationary:** the distribution π remains unchanged after a transition, that is,

$$\pi(\mathbf{x}') = \int_{\mathbb{R}^D} \pi(\mathbf{x}) T(\mathbf{x}, \mathbf{x}') d\mathbf{x}$$

2. **Ergodic:** meaning the chain converges to π in total variation from any initial distribution, $\pi^{(t)}(\mathbf{x}) \rightarrow \pi(\mathbf{x})$ as $t \rightarrow \infty$ for any $\pi^{(0)}(\mathbf{x})$

The chain is Ergodic if it is Harris positive recurrent and aperiodic, see Meyn and Tweedie [2012, Chapter 7].

The sampling methods we present satisfy the detailed balance condition. Detailed balance is a sufficient condition for the target distribution to be stationary. It states that the probability of being in state $\mathbf{x}$ and transitioning to state $\mathbf{x}'$ is the same as being in state $\mathbf{x}'$ and transitioning to state $\mathbf{x}$ and is defined as:

$$\pi(\mathbf{x}) T(\mathbf{x}, \mathbf{x}') = \pi(\mathbf{x}') T(\mathbf{x}', \mathbf{x}).$$

Next we describe the standard MCMC methods that are later extended, and briefly discuss geometric MCMC methods.

2.4.1 Slice sampling

Neal [2003] introduces slice sampling as a method that targets uniform sampling from the region under the graph of the probability density

$$\{(\mathbf{x}, s) \in \mathbb{R}^D \times \mathbb{R}_+ : 0 < s < p(\mathbf{x})\} \subset \mathbb{R}^{D+1}. \quad (2.14)$$

Uniform sampling from this set can be carried out by alternating between sampling $\mathbf{x}$ and s , and this procedure targets the correct distribution [Neal, 2003]. For a fixed level s , define the slice $L_{\mathbf{x}}(s) := \{\mathbf{x} \in \mathbb{R}^D : p(\mathbf{x}) > s\}$, one idealized iteration is:

1. Sample $s \sim \text{Unif}(0, p(\mathbf{x}))$.
2. Sample $\mathbf{x} \sim \text{Unif}(L_{\mathbf{x}}(s))$.

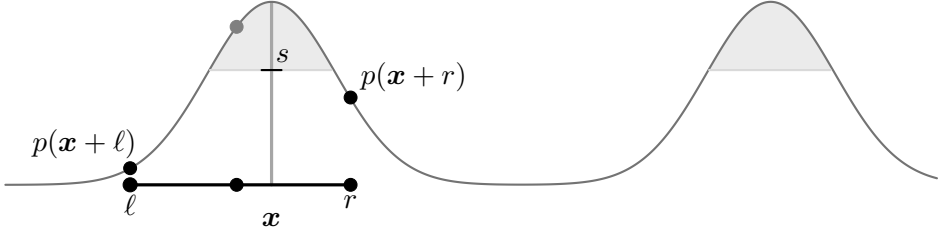


Figure 2.1: Visualization of the step-out procedure. The shaded gray area shows the region $L_{\mathbf{x}}(s)$, the step-out procedure took two steps to the left and one step to the right until stoppage. For a multimodal distribution the step-out procedure misses the second mode if the step-size w is not large enough.

Computing $L_{\mathbf{x}}(s)$ is possible, for example, in settings where the distribution is log-concave or rotationally invariant, in general we approximate the slice set with step-out and shrinkage procedures. We first describe the one-dimensional case and later the hit-and-run extension to higher dimensions. Assume the current state is $x \in \mathbb{R}$ and a value of s is drawn from $\text{Unif}(0, p(x))$ and define $\gamma_x(t) = x + t$.

Step-out procedure This procedure computes an open interval by expanding to the sides of the current position. Given two hyperparameters $w \in \mathbb{R}$ the expansion width and $m \geq 0$ the maximum number of expansions. Initialize for $u \sim \text{Unif}(0, w)$ $\ell = -u$ and $r = \ell + w$. Sample $\iota \sim \text{Unif}(\{1, \dots, m\})$.

- Right side: At most ι times expand as $r = r + w$ until the condition $p(\gamma_x(r)) < s$ is met.
- Left side: At most $m+1-\iota$ times expand $\ell = \ell - w$ until the condition $p(\gamma_x(\ell)) < s$ is met.

The interval $(x+\ell, x+r)$ is a wide approximation of $L_{\mathbf{x}}(s)$. A visualization of the step-out procedure is provided in Figure 2.1.

Shrinkage procedure Given the interval (ℓ, r) the shrinkage procedure draws a valid proposal by iteratively reducing the interval until a proposal falls in the intersection with the slice $L_{\mathbf{x}}(s)$. We follow the presentation of Durmus et al. [2026] which differs slightly from the original formulation of Neal [2003]. The procedure works as follows. Let $J = (\ell, r)$ and sample $a \sim \text{Unif}(J)$, if $\gamma_x(a) \in L_{\mathbf{x}}(s)$ the proposal is $\gamma_x(a) = x + a$. Sample

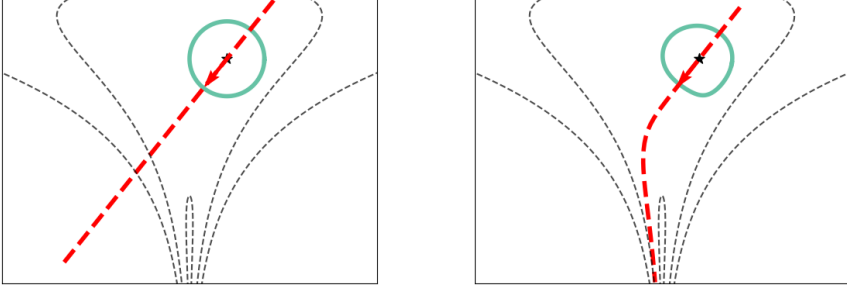


Figure 2.2: Visualization of the hit-and-run procedure. Left: A velocity uniformly sampled uniformly on $\mathbb{S}^{D-1}$ and the resulting straight line. Right: A velocity sampled uniformly from $\|\mathbf{v}\| : \|\mathbf{v}\|_{\mathbf{G}(\mathbf{x})}^2$ and the geodesic capture better the geometry of the target.

$b \sim \text{Unif}(J)$, if $\gamma_x(b) \in L_x(s)$ the proposal is $\gamma_x(b) = x + b$. Otherwise shrink the interval J by:

- Update J

$$J = \begin{cases} J \cap (a \wedge b, a \vee b), & \text{if } 0 \in J, \\ J \setminus (a \wedge b, a \vee b), & \text{if } 0 \notin J. \end{cases}$$

- Set $a \leftarrow b$ and sample $b \sim \text{Unif}(J)$.
- Repeat until $\gamma_x(b) \in L_x(s)$.

Hit-and-Run slice sampling There are multiple ways of extending slice sampling to multivariate distributions [Neal, 2003, Murray et al., 2010]. Here we consider the Hit-and-Run [Bélisle et al., 1993] extension which is relevant for our method. Denote the unit sphere by $\mathbb{S}^{D-1} := \{\mathbf{v} \in \mathbb{R}^D : \|\mathbf{v}\|^2 = 1\}$ (See Fig. 2.2). Given current position $\mathbf{x} \in \mathbb{R}^D$

- Sample direction $\mathbf{v} \sim \text{Unif}(\mathbb{S}^{D-1}(\mathbf{x}))$
- Define $\gamma_{(\mathbf{x}, \mathbf{v})}(t) = \mathbf{x} + t\mathbf{v}$ and apply one dimensional slice sampling on $t \mapsto \gamma_{(\mathbf{x}, \mathbf{v})}(t)$

Since the straight line $\gamma_{(\mathbf{x}, \mathbf{v})}(t)$ defines a univariate distribution, slice sampling is directly applicable. Concretely, one transition of slice sampling consist of the following steps: Sample $s \sim \text{Unif}(0, p(\mathbf{x}))$, the step-out procedure returns the interval

$$\ell, r = \text{Step-out}_{w,m}(s, \gamma_{(\mathbf{x}, \mathbf{v})}),$$

the shrinkage procedure gives the proposal

$$t^* = \text{Shrink}_{\ell,r}(s, \gamma(\mathbf{x}, \mathbf{v})).$$

Finally the new sample after one step of the algorithm is $\mathbf{x}' = \gamma(\mathbf{x}, \mathbf{v})(t^*)$. This allows sampling from target distributions defined on $\mathbb{R}^D$.

2.4.2 Geometric MCMC

By geometric MCMC we refer to MCMC methods that incorporate a Riemannian tensor in the sampling method. The main benefit is the adaptation to local curvature, but the introduction of a non-flat geometry results in higher computational cost. Recall that

$$\mathbf{G}(\cdot) : \mathbb{R}^D \rightarrow \mathbb{R}^D \times \mathbb{R}^D$$

is the matrix that holds the components of a metric tensor (See Section 2.1). Next we describe some geometric MCMC methods and their computational cost.

Riemannian manifold Hamiltonian Monte Carlo (RMHMC) The seminal work of Girolami and Calderhead [2011] extends two gradient-based sampling methods, Metropolis-adjusted Langevin algorithm (MALA) and Hamiltonian Monte Carlo (HMC) from Euclidean space to a Riemannian setting. Here we only describe the HMC extension. Both HMC and RMHMC introduce an auxiliary momentum $\mathbf{p} \in \mathbb{R}^D$ and model a system of ordinary differential equations (ODEs). The goal is that the solution to the ODEs generates samples that are further away from the initial state reducing sample correlation. The Hamiltonian equation is

$$\mathcal{H}(\mathbf{x}, \mathbf{p}) = \phi(\mathbf{x}) + \frac{1}{2} \mathbf{p}^\top \mathbf{G}(\mathbf{x})^{-1} \mathbf{p},$$

where $\phi(\mathbf{x}) = -\ell(\mathbf{x}) + \frac{1}{2} \log \det \mathbf{G}(\mathbf{x})$. The dynamics are given by the differential equations obtained from differentiation of $\mathcal{H}(\mathbf{x}, \mathbf{p})$

$$\dot{\mathbf{x}} = \mathbf{G}(\mathbf{x})^{-1} \mathbf{p}, \quad \dot{\mathbf{p}} = -\nabla_{\mathbf{x}} \mathcal{H}(\mathbf{x}, \mathbf{p}), \quad (2.15)$$

The numerical integration of Eq. 2.15 relies on fixed point iteration [Girolami and Calderhead, 2011]. Each step iterates until a specified tolerance is reached to generate a proposal, increasing the runtime. With a full rank metric tensor the resulting computations are of order $\mathcal{O}(D^3)$ at each iteration, which limits its scalability to high dimensional problems.

Lagrangian dynamical Monte Carlo (LMC) Lan et al. [2015] expresses the Hamiltonian equations in terms of velocity instead of momentum through a change of variables $\mathbf{v} = \mathbf{G}(\mathbf{x})^{-1}\mathbf{p}$, and the system dynamics become

$$\dot{\mathbf{x}} = \mathbf{v}, \quad \dot{\mathbf{v}} = -\boldsymbol{\eta}(\mathbf{x}, \mathbf{v}) - \mathbf{G}(\mathbf{x})^{-1}\nabla_{\mathbf{x}}\phi(\mathbf{x}), \quad (2.16)$$

where $[\boldsymbol{\eta}(\mathbf{x}, \mathbf{v})]_k = \sum_{i,j} \Gamma_{ij}^k(\mathbf{x}) v_i v_j$, and the Christoffel symbols Γ_{ij}^k (see Eq. 2.3). The authors propose a numerical integrator that is explicit and does not rely on fixed point iterations improving the running time over RMHMC. Each step still computes inverses of the metric tensor and is $\mathcal{O}(D^3)$ in complexity which again limits its scalability to higher dimensions. One way of reducing the complexity of the numerical integrators are metrics with closed form inverse and determinant [Hartmann et al., 2022]. Some metric choices are presented in Chapter 3. Additionally, the numerical integrator introduces two hyperparameters, the step-size and the number of steps, with several strategies available for selecting a value. Williams et al. [2024] discuss the choice of metric, numerical integrator, and hyperparameter adaptation in RMHMC and LMC.

2.5 Modeling discrete data

We study two problems: discrete-data generation and classification. For generative modeling we consider flow matching and Riemannian flow matching, while for classification we use Gaussian processes. We first review flow-based models and a Riemannian flow matching approach used for generative modeling of discrete data, and then we review Gaussian process classification. Both approaches model a continuous latent process that produces discrete outputs via an argmax operation. In the first case the latent model is flow based, and in the second it is a Gaussian process.

2.5.1 Discrete data generation

We first two continuous generative modes that are building blocks: conditional flow matching which operates in the Euclidean space, then Riemannian flow matching which operates on manifolds. Then we present a method for discrete data generation which operates on the simplex and used the Fisher geometry.

Conditional flow matching (CFM) Conditional flow matching is a generative model for continuous data in the Euclidean space [Lipman et al., 2023, Albergo and Vanden-Eijnden, 2023, Liu et al., 2023]. For $\mathbf{x} \in \mathbb{R}^D$,

the model defines a vector field $\mathbf{u}_t : \mathbb{R}^D \times [0, 1] \rightarrow \mathbb{R}^D$, which evolves data according to

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{u}_t(\mathbf{x}_t), \quad \mathbf{x}_t \sim p_t, \quad (2.17)$$

where the initial condition is a simple base distribution p_0 that allows direct generation of samples and density evaluation, and the final condition is the data distribution $p_1 = p_{\text{data}}$. The marginal probability path $\{p_t\}_{t \in [0, 1]}$ is linked to $\mathbf{u}_t$ by the continuity equation, this gives the instantaneous change of variable formula, which is useful for computing log-densities

$$\frac{d \log p_t(\mathbf{x}_t)}{dt} = -\nabla \cdot (\mathbf{u}_t(\mathbf{x}_t)).$$

In practice $\mathbf{u}_t$ is not available, instead a parametric model (usually a neural network) $\mathbf{v}_t^\theta$ is fit to the conditional vector fields $\mathbf{u}_t(\mathbf{x}_t \mid \mathbf{x}_0, \mathbf{x}_1)$, the resulting objective is

$$\mathcal{L}_{\text{CFM}}(\theta) = \mathbb{E}_{\substack{t \sim \text{Unif}(0, 1) \\ \mathbf{x}_0, \mathbf{x}_1 \sim \pi(\mathbf{x}_0, \mathbf{x}_1)}} \left\| \mathbf{v}_t^\theta(\mathbf{x}_t) - \mathbf{u}_t(\mathbf{x}_t \mid \mathbf{x}_0, \mathbf{x}_1) \right\|_2^2, \quad (2.18)$$

where π is any coupling with marginals p_0 and p_1 . Lipman et al. [2023] shows that this conditional objective equals up to a constant, the objective for estimating $\mathbf{u}_t(\mathbf{x}_t)$ directly. Therefore, $\mathbf{v}_t^\theta(\mathbf{x}_t)$ is used in Equation 2.17 to generate new data. The velocity field depends on the interpolation path between $\mathbf{x}_0$ and $\mathbf{x}_1$. A common choice is the linear interpolation $\mathbf{x}_t = (1 - t)\mathbf{x}_0 + t\mathbf{x}_1$ which yields the constant conditional velocity

$$\mathbf{u}_t(\mathbf{x}_t \mid \mathbf{x}_0, \mathbf{x}_1) = \dot{\mathbf{x}}_t = \mathbf{x}_1 - \mathbf{x}_0.$$

Two common options for the joint density of $\mathbf{x}_0, \mathbf{x}_1$ are the independent coupling $\pi(\mathbf{x}_0, \mathbf{x}_1) = p_0(\mathbf{x}_0)p_1(\mathbf{x}_1)$ and optimal-transport coupling. The optimal transport coupling is approximated with mini-batch optimal transport and typically improves optimization stability while reducing the number of function evaluations needed at sampling time [Tong et al., 2024].

Riemannian conditional flow matching (RCFM) Conditional flow matching can be extended from Euclidean space to a Riemannian manifold $(\mathcal{M}, \mathbf{G})$ [Chen and Lipman, 2024]. Instead of a vector field in the Euclidean space $\mathbb{R}^D$, the vector field is defined on the tangent space of the manifold $\mathbf{u}_t(\mathbf{x}) \in T_{\mathbf{x}}\mathcal{M}$. The resulting dynamics on the manifold are

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{u}_t(\mathbf{x}_t), \quad \mathbf{x}_t \sim p_t.$$

where the initial condition is a simple base distribution defined over $\mathcal{M}$. The marginal probability paths are generated from $\mathbf{u}_t$ by the Riemannian version of the instantaneous change of variables

$$\frac{d \log p_t(\mathbf{x}_t)}{dt} = -\operatorname{div}_{\mathbf{G}}(\mathbf{u}_t(\mathbf{x}_t)),$$

where $\operatorname{div}_{\mathbf{G}}$ is the Riemannian divergence [Lee, 2018]. Similarly to the Euclidean case, $\mathbf{u}_t$ is not available, but we can train the parametric model $\mathbf{v}_t^\theta$ using the conditional objective

$$\mathbb{E}_{t \sim \operatorname{Unif}(0,1)} \left\| \mathbf{v}_t^\theta(\mathbf{x}_t) - \mathbf{u}_t(\mathbf{x}_t \mid \mathbf{x}_0, \mathbf{x}_1) \right\|_{\mathbf{G}(\mathbf{x}_t)}^2.$$

Here $\|\mathbf{v}\|_{\mathbf{G}(\mathbf{x}_t)} := \sqrt{\mathbf{v}^\top \mathbf{G}(\mathbf{x}_t) \mathbf{v}}$ is the norm induced by the metric $\mathbf{G}$ at $\mathbf{x}_t$. The tangent velocity field depends on the interpolation, a standard choice is the geodesic which connects $\mathbf{x}_0$ and $\mathbf{x}_1$, defined by $\mathbf{x}_t = \operatorname{Exp}_{\mathbf{x}_0}(t\mathbf{v})$ and $\mathbf{v} = \operatorname{Log}_{\mathbf{x}_0}(\mathbf{x}_1)$. This choice gives the conditional target velocity

$$\mathbf{u}_t(\mathbf{x}_t \mid \mathbf{x}_0, \mathbf{x}_1) = \frac{\operatorname{Log}_{\mathbf{x}_t}(\mathbf{x}_1)}{1-t}, \quad \mathbf{x}_t = \operatorname{Exp}_{\mathbf{x}_0}(t \operatorname{Log}_{\mathbf{x}_0}(\mathbf{x}_1)).$$

When $\mathcal{M} = \mathbb{R}^D$ with the Euclidean metric, these expressions reduce to standard CFM. If the manifold does not have closed form geodesics and logarithm map, then training and inference rely on approximations of these quantities. As in CFM, $\pi(\mathbf{x}_0, \mathbf{x}_1)$ can be the independent coupling or the optimal transport coupling. On the manifold, the geodesic distance of two points is the length of the geodesic between them, mini-batch optimal transport computes the transport cost with respect to this distance.

Discrete data generation with the Fisher geometry Cheng et al. [2024] and Davis et al. [2024] introduce Categorical Flow Matching on Statistical Manifolds (SFM), a special case of Riemannian flow matching for discrete data. In this model the manifold is the simplex and the metric is the Fisher information metric. In practice, the flow could be modeled directly on the simplex, but computations become unstable near the boundaries, where discrete observations lie. SFM therefore models Riemannian flow matching on the sphere instead. We first introduce the simplex geometry with the Fisher metric, then establish the isomorphism with the sphere, and finally specify the training and sampling procedures.

Let K be the number of categories and $D := K - 1$ the latent dimension. Consider the simplex $\Delta^D := \{\mathbf{x} \in \mathbb{R}_{\geq 0}^K : \sum_{k=1}^K x_k = 1\}$, and endow the

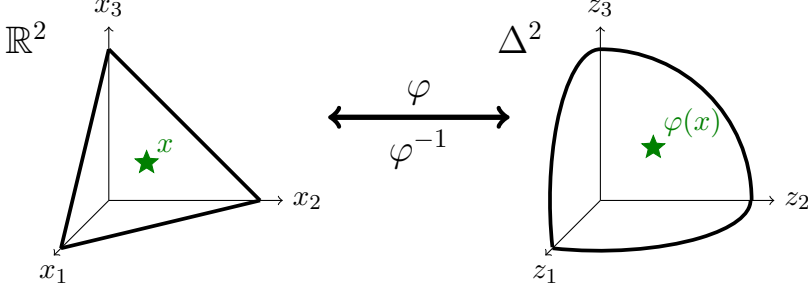


Figure 2.3: Visual illustration of the simplex-to-sphere map.

space with the Fisher information metric of the categorical distribution [Miyamoto et al., 2024]:

$$\mathbf{G}_F(\mathbf{x}) = \text{diag}\left(\frac{1}{\mathbf{x}_{1:D}}\right) + \frac{1}{x_K} \mathbf{1}\mathbf{1}^\top.$$

Under the Fisher metric, the space has the inner product for $\mathbf{u}, \mathbf{v} \in T_{\mathbf{x}}\Delta^D$,

$$\langle \mathbf{u}, \mathbf{v} \rangle_{G_F(\mathbf{x})} = \sum_{i=1}^D \frac{u_i v_i}{x_i} + \frac{(\sum_{i=1}^D u_i)(\sum_{i=1}^D v_i)}{x_K} = \sum_{i=1}^K \frac{u_i v_i}{x_i}.$$

The last equality uses the tangent space constraint in the ambient space $T_{\mathbf{x}}\Delta^D = \{\mathbf{u} \in \mathbb{R}^K : \sum_{i=1}^K u_i = 0\}$ so $u_K = -\sum_{i=1}^D u_i$. Note that the inner product is only defined on $\mathring{\Delta}^D$, since it is undefined on the boundary. The following bijection connects the simplex to the positive orthant of the sphere:

$$\begin{aligned} \varphi : \Delta^D &\rightarrow \mathbb{S}_+^D, & \mathbf{x} &\mapsto \mathbf{z} = \sqrt{\mathbf{x}}, \\ \varphi^{-1} : \mathbb{S}_+^D &\rightarrow \Delta^D, & \mathbf{z} &\mapsto \mathbf{x} = \mathbf{z}^2. \end{aligned} \tag{2.19}$$

where $\mathbb{S}_+^D := \{\mathbf{z} \in \mathbb{R}_{\geq 0}^K : \sum_{k=1}^K z_k^2 = 1\}$. This bijection (illustrated in Fig. 2.3) is in fact an isomorphism between $(\Delta^D, \mathbf{G}_F)$ and $(\mathbb{S}_+^D, \mathbf{G})$, where $\mathbf{G}$ is the canonical metric on the sphere [Miyamoto et al., 2024]. The sphere is embedded in $\mathbb{R}^K$, and it inherits the ambient space inner product restricted to the tangent space, for $\mathbf{u}, \mathbf{v} \in T_{\mathbf{z}}\mathbb{S}_+^D$,

$$\langle \mathbf{u}, \mathbf{v} \rangle_G = \langle \mathbf{u}, \mathbf{v} \rangle_2 = \sum_{i=1}^K u_i v_i.$$

The geodesics in the sphere are the outer rings, the exponential and logarithm map are:

$$\begin{aligned}\text{Exp}_{\mathbf{z}}(t\mathbf{v}) &= \cos(t\|\mathbf{v}\|_2)\mathbf{z} + \sin(t\|\mathbf{v}\|_2)\frac{\mathbf{v}}{\|\mathbf{v}\|_2}, \\ \text{Log}_{\mathbf{z}}(\mathbf{z}') &= \frac{\theta}{\sin\theta}(\mathbf{z}' - \cos(\theta)\mathbf{z}), \quad \theta = \arccos(\langle \mathbf{z}, \mathbf{z}' \rangle).\end{aligned}\tag{2.20}$$

Proposition 1 (Isometry, [Amari and Nagaoka, 2000, Miyamoto et al., 2024]). *The diffeomorphism in Equation 2.19 is an isomorphism between $(\Delta^D, \mathbf{G}_F)$ and $(\mathbb{S}_+^D, \mathbf{G})$ up to a factor of $\frac{1}{4}$.*

Proof. The proof consists on showing that $\frac{1}{4}\langle \mathbf{u}, \mathbf{v} \rangle_{\mathbf{G}_F(\mathbf{x})} = \langle d\varphi_{\mathbf{x}}(\mathbf{u}), d\varphi_{\mathbf{x}}(\mathbf{v}) \rangle$ for all $\mathbf{x} \in \Delta^D$ and $\mathbf{u}, \mathbf{v} \in T_{\mathbf{x}}\Delta^D$, where $d\varphi_{\mathbf{x}}(\mathbf{v})$ denotes the differential of φ at $\mathbf{x}$ applied to the tangent vector $\mathbf{v}$. Since $\mathbb{S}_+^D$ is embedded in $\mathbb{R}^K$, we use the ambient Euclidean inner product

$$\begin{aligned}\langle d\varphi_{\mathbf{x}}(\mathbf{u}), d\varphi_{\mathbf{x}}(\mathbf{v}) \rangle &= \left\langle \left. \frac{d}{dt}\varphi(\mathbf{x} + t\mathbf{u}) \right|_{t=0}, \left. \frac{d}{dt}\varphi(\mathbf{x} + t\mathbf{v}) \right|_{t=0} \right\rangle \\ &= \left\langle \left(\frac{u_1}{2\sqrt{x_1}}, \dots, \frac{u_K}{2\sqrt{x_K}} \right), \left(\frac{v_1}{2\sqrt{x_1}}, \dots, \frac{v_K}{2\sqrt{x_K}} \right) \right\rangle \\ &= \frac{1}{4} \sum_{i=1}^K \frac{u_i v_i}{x_i} = \frac{1}{4} \langle \mathbf{u}, \mathbf{v} \rangle_{\mathbf{G}_F(\mathbf{x})}.\end{aligned}$$

□

Based on the isomorphism (Prop. 1), the exponential map in $(\Delta^D, \mathbf{G}_F)$ is

$$\begin{aligned}\text{Exp}_{\mathbf{x}}(t\mathbf{v}) &= \varphi^{-1}(\text{Exp}_{\sqrt{\mathbf{x}}}(d\varphi_{\mathbf{x}}(\mathbf{v}))) \\ &= \left(\cos\left(\frac{t\|\mathbf{v}\|_{\mathbf{G}_F(\mathbf{x})}}{2}\right) \sqrt{\mathbf{x}} + \sin\left(\frac{t\|\mathbf{v}\|_{\mathbf{G}_F(\mathbf{x})}}{2}\right) \frac{\mathbf{v}}{\|\mathbf{v}\|_{\mathbf{G}_F(\mathbf{x})\sqrt{\mathbf{x}}}} \right)^2\end{aligned}$$

Note that $\|\cdot\|_{\mathbf{G}_F}$ and $\frac{1}{\sqrt{\mathbf{x}}}$ explode when $\mathbf{x}$ approaches $\partial\Delta^D$ which makes it unstable, to address this Cheng et al. [2024] and Davis et al. [2024] proposed modeling directly in the sphere. The approach first transforms the training data to the sphere with φ , and then fits the flow’s parameters with RCFM on this space (Algorithm 1). New data is sampled by numerical integration of the learned velocity field, and the final point is mapped to the simplex with φ^{-1} (Algorithm 2).

Algorithm 1 Training of SFM

Require: Data $\mathbf{x} \in \Delta^D$, bijection φ , base p_0 (e.g. $\varphi_{\#}\text{Unif}(\Delta^D)$), coupling π (indep. or minibatch OT)

- 1: **for** each mini-batch **do**
 - 2: $\mathbf{z}_1 \leftarrow \varphi(\mathbf{x})$ $\triangleright$ To sphere
 - 3: Sample $\mathbf{z}_0 \sim p_0$; pair $(\mathbf{z}_0, \mathbf{z}_1)$ via π
 - 4: Sample $t \sim \text{Unif}(0, 1)$
 - 5: $\mathbf{z}_t \leftarrow \text{Exp}_{\mathbf{z}_0}(t \text{Log}_{\mathbf{z}_0}(\mathbf{z}_1))$, $\mathbf{u}_t \leftarrow \text{Log}_{\mathbf{z}_t}(\mathbf{z}_1)/(1 - t)$
 - 6: Update θ by $\min \|\mathbf{v}_{\theta}(\mathbf{z}_t, t) - \mathbf{u}_t\|_{\mathbf{G}(\mathbf{z}_t)}^2$
-

Algorithm 2 Sampling with SFM

Require: Learned $\mathbf{v}_{\theta}$, bijection φ , base p_0

- 1: Sample $\mathbf{z}_0 \sim p_0$
 - 2: Solve ODE: $\frac{d\mathbf{z}_t}{dt} = \mathbf{v}_{\theta}(\mathbf{z}_t, t)$, $t \in [0, 1]$, $\mathbf{z}_t \in \mathbb{S}^D$
 - 3: $\hat{\mathbf{x}} \leftarrow \varphi^{-1}(\mathbf{z}_1)$
-

2.5.2 Gaussian processes for classification

We briefly introduce Gaussian processes and present two alternatives for adapting Gaussian processes for classification as background for Paper IV.

Gaussian process A Gaussian process (GP) [Williams and Rasmussen, 2006] defines a distribution over functions such that any finite collection of function values is jointly Gaussian. It provides a flexible prior together with uncertainty quantification. Given a finite set of inputs $\mathbf{x}_1, \dots, \mathbf{x}_N \in \mathbb{R}^P$ the distribution over values $(f(\mathbf{x}_1), \dots, f(\mathbf{x}_N))$ is multivariate Gaussian, where the mean is specified by a mean function $m : \mathbb{R}^P \rightarrow \mathbb{R}$ and the covariance between any pair of points by a kernel $K_{\theta} : \mathbb{R}^P \times \mathbb{R}^P \rightarrow \mathbb{R}$ dependent on hyperparameters θ . The GP prior is written as

$$f \sim \mathcal{GP}(m(\cdot), K_{\theta}(\cdot, \cdot)).$$

Consider a supervised regression problem where the data is denoted $\mathcal{D} := \{(\mathbf{x}_n, \mathbf{z}_n)\}_{n=1}^N$ for $\mathbf{x}_n \in \mathbb{R}^P$ and $\mathbf{z}_n \in \mathbb{R}^D$. In practice, the responses are usually centered by subtracting the empirical mean, and the mean function is set to zero. The kernel function imposes assumptions on the correlation of the latent functions and on the behavior of the function space. For example, different kernels can model periodic behaviors, or be translation

invariant. In this work we consider the radial basis function kernel

$$K_\theta(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|_2^2}{2\ell^2}\right), \quad \theta = (\sigma_f^2, \ell),$$

where σ_f^2 controls the variance and ℓ is the length scale which controls how fast the dependency between $\mathbf{x}$ and $\mathbf{x}'$ decays. This kernel gives smooth latent functions.

Although it is possible to introduce dependency among dimensions [Goulard and Voltz, 1992], in this thesis, we model D independent GPs (one per dimension) denoted $\mathbf{f} = (f^{(1)}, \dots, f^{(D)})$, and the same kernel is shared across D -dimensions. Given a GP prior assume the noisy Gaussian likelihood

$$\mathbf{z}_n \mid \mathbf{f}(\mathbf{x}_n) \sim \mathcal{N}(\mathbf{f}(\mathbf{x}_n), \sigma^2 \mathbf{I}_D).$$

Let $\mathbf{x}_* \in \mathbb{R}^P$ be a test input and denote the vector of responses $\mathbf{z}^{(d)} := (z_1^{(d)}, \dots, z_N^{(d)})^\top \in \mathbb{R}^N$ for $d = 1, \dots, D$. The joint distribution of $\mathbf{z}^{(d)}$ and $f^{(d)}(\mathbf{x}_*)$ is:

$$\begin{bmatrix} \mathbf{z}^{(d)} \\ f^{(d)}(\mathbf{x}_*) \end{bmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{bmatrix} \mathbf{K}_N + \sigma^2 \mathbf{I}_N & \mathbf{k}_* \\ \mathbf{k}_*^\top & k_{**} \end{bmatrix}\right),$$

where the covariance entries are given by $[\mathbf{K}_N]_{ij} = k_\theta(\mathbf{x}_i, \mathbf{x}_j)$ $[\mathbf{k}_*]_i = k_\theta(\mathbf{x}_i, \mathbf{x}_*)$, and $k_{**} = k_\theta(\mathbf{x}_*, \mathbf{x}_*)$. The standard predictive posterior is obtained by conditioning on the observations [Williams and Rasmussen, 2006], it is distributed as $f^{(d)}(\mathbf{x}_*) \mid \mathbf{z}^{(d)} \sim \mathcal{N}(\mu_*^{(d)}, \sigma_*^2)$ with mean and covariance

$$\begin{aligned} \mu_*^{(d)} &:= \mathbf{k}_*^\top (\mathbf{K}_N + \sigma^2 \mathbf{I}_N)^{-1} \mathbf{z}^{(d)}, \\ \sigma_*^2 &:= k_{**} - \mathbf{k}_*^\top (\mathbf{K}_N + \sigma^2 \mathbf{I}_N)^{-1} \mathbf{k}_*. \end{aligned} \tag{2.21}$$

We can write it compactly in a single expression

$$\mathbf{f}_* \mid \mathcal{D} \sim \mathcal{N}(\boldsymbol{\mu}_*, \boldsymbol{\Sigma}_*)$$

with $\boldsymbol{\mu}_* := (\mu_*^{(1)}, \dots, \mu_*^{(D)})^\top$ and $\boldsymbol{\Sigma}_* := \sigma_*^2 \mathbf{I}_D$. The marginal log-likelihood is used to learn the kernel hyperparameters. Since the likelihood is Gaussian and the output dimensions are assumed independent, it takes the form

$$\log p(\mathbf{z} \mid \theta) = -\frac{1}{2} \sum_{d=1}^D \|\mathbf{z}^{(d)}\|_{(\mathbf{K}_N + \sigma^2 \mathbf{I}_N)^{-1}}^2 - \frac{D}{2} \log |\mathbf{K}_N + \sigma^2 \mathbf{I}_N|, \tag{2.22}$$

where the constant terms that do not depend on θ are dropped.

Sparse Gaussian process Exact GP inference scales cubically in complexity with the number of training points, which prohibits its use with data points in the thousands. Sparse GP methods reduce this cost by using a smaller set $M \ll N$ of inducing points $\mathbf{X}_u := \{\mathbf{x}_m^u\}_{m=1}^M$. Denote the inducing variables for each dimension by

$$\mathbf{u}^{(d)} := f^{(d)}(\mathbf{X}_u) \in \mathbb{R}^M.$$

We first introduce the *uncollapsed* variational formulation [Hensman et al., 2013], and then the *collapsed* formulation [Titsias, 2009]. These methods use a variational Gaussian distribution

$$q(\mathbf{u}) = \prod_{d=1}^D q(\mathbf{u}^{(d)}), \quad q(\mathbf{u}^{(d)}) = \mathcal{N}(\mathbf{u}^{(d)} \mid \mathbf{m}^{(d)}, \mathbf{S}^{(d)}),$$

and define the approximation to the process

$$q(\mathbf{f}^{(d)}) = \int p(\mathbf{f}^{(d)} \mid \mathbf{u}^{(d)}) q(\mathbf{u}^{(d)}) d\mathbf{u}^{(d)}.$$

Similar to the derivation of Equation 2.13, the approximate process results in the following lower bound

$$\log p(\mathbf{z}) \geq \sum_{d=1}^D \mathbb{E}_{q(\mathbf{f}^{(d)})} \left[\log p(\mathbf{z}^{(d)} \mid \mathbf{f}^{(d)}) \right] - \sum_{d=1}^D \text{KL}[q(\mathbf{u}^{(d)}) \parallel p(\mathbf{u}^{(d)})]. \quad (2.23)$$

For the Gaussian approximation, the expected log likelihood factorizes over the data points

$$\log p(\mathbf{z}) \geq \sum_{d=1}^D \sum_{n=1}^N \mathbb{E}_{q(f_n^{(d)})} \left[\log \mathcal{N}(z_n^{(d)} \mid f_n^{(d)}, \sigma^2) \right] - \sum_{d=1}^D \text{KL}[q(\mathbf{u}^{(d)}) \parallel p(\mathbf{u}^{(d)})].$$

This is the *uncollapsed* bound. Since the first term is additive over the data it is possible to use stochastic optimization giving a complexity of $\mathcal{O}(BM^2 + M^3)$ for a batch size B . The variational parameters $\{\mathbf{m}^{(d)}, \mathbf{S}^{(d)}\}_{d=1}^D$ and the inducing points are optimized together.

The *collapsed* bound is obtained observing, following Titsias [2009], that the variational parameters $\{\mathbf{m}^{(d)}, \mathbf{S}^{(d)}\}_{d=1}^D$ can be optimized analytically. This leads to a bound that only depends on the inducing point locations and is not additive over the input data

$$\log p(\mathbf{z}) \geq \sum_{d=1}^D \log \mathcal{N}(\mathbf{z}^{(d)} \mid \mathbf{0}, \mathbf{Q}_N + \sigma^2 \mathbf{I}_N) - \frac{D}{2\sigma^2} \text{tr}(\mathbf{K}_N - \mathbf{Q}_N), \quad (2.24)$$

where $\mathbf{Q}_N = \mathbf{K}_{NM} \mathbf{K}_M^{-1} \mathbf{K}_{MN}$ and $\mathbf{K}_{NM}$ denotes the covariance between training and inducing inputs. The first term is the log-likelihood under the Nyström approximation $\mathbf{Q}_N$, while the second term measures the discrepancy between $\mathbf{K}_N$ and $\mathbf{Q}_N$, the complexity of the collapsed method is $\mathcal{O}(NM^2)$. If the inducing points are the training points then $\mathbf{K}_N = \mathbf{Q}_N$ and the objective is exact and coincides with the marginal log-likelihood.

The predictions over a new data point $\mathbf{x}_*$ are obtained by the predictive distribution given by the inducing variables

$$q(f_*^{(d)}) = \int p(f_*^{(d)} | \mathbf{u}^{(d)}) q(\mathbf{u}^{(d)}) d\mathbf{u}^{(d)},$$

which has a closed form because $q(\mathbf{u}^{(d)})$ is Gaussian. In the *uncollapsed* case the parameters of the variational Gaussian are optimized with respect to Equation 2.23. In the *collapsed* case the parameters are given analytically as solutions from Equation 2.24. The complexity of the predictions is cubic on M .

Dirichlet-based Gaussian process classification Gaussian process classification estimates class probabilities from latent Gaussian processes. Let us consider classification with K categories where the data is $\mathcal{D} = \{(\mathbf{x}_n, c_n)\}_{n=1}^N$ for class $c_n \in \{1, \dots, K\}$ and covariate $\mathbf{x}_n \in \mathbb{R}^P$. The class probabilities are denoted $\boldsymbol{\pi}(\mathbf{x}) \in \Delta^D$. A standard GP classifier introduces K latent Gaussian processes, giving the model

$$f_k(\cdot) \sim \mathcal{GP}(0, K_\theta(\cdot, \cdot)), \quad c|\mathbf{x} \sim \text{Cat}(\text{softmax}(\mathbf{f}(\mathbf{x}))),$$

where $\mathbf{f}(\mathbf{x}) = (f^{(1)}(\mathbf{x}), \dots, f^{(K)}(\mathbf{x}))^\top$. The categorical likelihood is not conjugate with the Gaussian prior and posterior inference relies on approximations, such as variational inference [Hensman et al., 2015].

Dirichlet-based Gaussian process classification (GPD) [Milios et al., 2018] avoids this non-conjugacy by converting classification to a regression problem in a latent space. Instead of modeling the class probabilities directly with a link function (e.g. softmax), it places a Dirichlet distribution on the probabilities

$$\boldsymbol{\pi}(\mathbf{x}) | \mathbf{x} \sim \text{Dir}(\boldsymbol{\alpha}(\mathbf{x})).$$

The Dirichlet parameters are modeled with GPs. Conditional on a label $c = k$, the parameters of the Dirichlet distribution are chosen as

$$\alpha_k = \mathbf{1}\{c = k\} + \alpha_\varepsilon, \quad k = 1, \dots, K,$$

where $\alpha_\varepsilon \geq 0$ is a smoothing parameter such that as $\alpha_\varepsilon \rightarrow 0$ the Dirichlet distribution concentrates on the one-hot vector $\mathbf{e}_k$. The distribution is

generated by K -independent Gamma random variables distributed $g_j \stackrel{\text{ind}}{\sim} \text{Gamma}(\alpha_j, 1)$ and $\boldsymbol{\pi}$ is recovered by normalization

$$\pi_k = \frac{g_k}{\sum_{\ell=1}^K g_\ell} \iff \boldsymbol{\pi} = \text{softmax}(\log \mathbf{g}).$$

To make the model conjugate, GPD approximates each Gamma distribution with a log-normal random variable $\text{Lognormal}(\tilde{y}_j, \tilde{\sigma}_j^2)$, matching the first two moments

$$\tilde{\sigma}_k^2 = \log\left(1 + \frac{1}{\alpha_k}\right), \quad \tilde{y}_k = \log \alpha_k - \frac{1}{2}\tilde{\sigma}_k^2. \quad (2.25)$$

Under the log-normal approximation the model is conjugate in log space. The method fits K -independent GPs to the transformed data $\tilde{y}_{nj}$ with heteroscedastic noise variance $\tilde{\sigma}_{nj}^2$, where both $\tilde{y}_{nj}$ and $\tilde{\sigma}_{nj}^2$ depend on the class label and the smoothing parameter. At a test input $\mathbf{x}_*$, inference for the latent variable $\mathbf{f}_*$ is exact, but class probabilities are computed after the softmax map and are estimated with Monte Carlo

$$\mathbb{E}[\boldsymbol{\pi}_* \mid \mathbf{x}_*, \mathcal{D}] \approx \frac{1}{S} \sum_{s=1}^S \text{softmax}(\mathbf{f}_*^{(s)}),$$

$$\mathbf{f}_*^{(s)} \sim p(\mathbf{f}_* \mid \mathbf{x}_*, \mathcal{D}).$$

Note that since GP classification is converted to GP regression on the smoothed labels $\tilde{y}$, it can directly use sparse Gaussian processes or any other scalable regression machinery.

Chapter 3

Advances in Geometric Bayesian Inference

In this chapter we propose using the geometry of the log target density in two different inference methods for drawing samples. The first one (Paper I) is a distributional approximation to the target distribution using a generalization of the Laplace approximation. The second one (Paper II) uses MCMC and generalizes the hit-and-run slice-sampler to a particular Riemannian setting in which geodesics are not closed form.

The Chapter starts by introducing common elements between our contributions. These elements rely on the notions Riemannian geometry from Chapter 2. We introduce choices for the metric tensor, the volume, the Hausdorff density, and discuss the numerical computation of the exponential map. Then, we present the Riemannian Laplace approximation with the Fisher metric and the geodesic slice sampler for multimodal distributions with strong curvature.

Motivation Equipping the space with a non Euclidean geometry increases computational complexity (dependent on the metric choice), but can lead to higher quality samples. In particular, it enables exploration of curved regions of the parameter space and better captures multimodal distributions compared to Euclidean methods.

3.1 Choice of metric tensor

We consider a target density $p(\mathbf{x})$ defined over $\mathbb{R}^D$ with respect to the Lebesgue measure. We give the Euclidean space the Riemannian metric $\mathbf{G}(\mathbf{x})$. Once the metric is fixed it determines the local geometry of the

space and determines the geometric objects used in this chapter such as geodesics, logarithm map and volume elements.

3.1.1 Riemannian metrics

The Riemannian metrics we consider in this chapter depend on the probabilistic model, through the log density $\log p(\mathbf{x})$, and each of them leads to a different geometry. Next we introduce the metric choices, and give the corresponding inverse, determinant and Christoffel symbols needed geodesic computations.

Euclidean metric Let us present Euclidean space using the same tools as more general cases. Euclidean space is equipped with a constant metric,

$$\mathbf{G}(\mathbf{x}) = \mathbf{I}_D.$$

The Christoffel symbols vanish $\Gamma_{ij}^k = 0$ for all $i, j, k = 1, \dots, D$ and the exponential and logarithm map are, for $\mathbf{x}, \mathbf{y} \in \mathbb{R}^D$ and $\mathbf{v} \in T_{\mathbf{x}}\mathbb{R}^D \cong \mathbb{R}^D$,

$$\begin{aligned} \text{Exp}_{\mathbf{x}}(t\mathbf{v}) &= \mathbf{x} + t\mathbf{v}, \\ \text{Log}_{\mathbf{x}}(\mathbf{y}) &= \mathbf{y} - \mathbf{x} \end{aligned} \tag{3.1}$$

The volume measure is the usual Lebesgue measure, geodesics are straight lines and all operations are available in closed form. The inverse metric $\mathbf{G}^{-1}$ is again the identity, and the determinant is one. In other words, we obtain the usual properties of the Euclidean metric.

Fisher metric Let us consider a probabilistic model such that target density is $p_{\mathcal{Y}|\mathcal{X}}(\mathbf{y} | \mathbf{x}) p_{\mathcal{X}}(\mathbf{x})$. The Fisher information matrix [Fisher, 1925, Amari and Nagaoka, 2000] is defined as

$$\mathbf{G}(\mathbf{x}) = \mathbb{E}_{\mathbf{y}|\mathbf{x}} \left[\nabla_{\mathbf{x}} \log p_{\mathcal{Y}|\mathcal{X}}(\mathbf{y} | \mathbf{x}) \nabla_{\mathbf{x}} \log p_{\mathcal{Y}|\mathcal{X}}(\mathbf{y} | \mathbf{x})^{\top} \right]. \tag{3.2}$$

The Fisher information can be motivated by the local approximation to the Kullback-Leibler divergence between two nearby distributions. Consider the models at $\mathbf{x}$ and $\mathbf{x} + d\mathbf{x}$, a second order Taylor expansion gives

$$\begin{aligned} \log p_{\mathcal{Y}|\mathcal{X}}(\mathbf{y} | \mathbf{x} + d\mathbf{x}) &\approx \log p_{\mathcal{Y}|\mathcal{X}}(\mathbf{y} | \mathbf{x}) + \langle d\mathbf{x}, \nabla_{\mathbf{x}} \log p_{\mathcal{Y}|\mathcal{X}}(\mathbf{y} | \mathbf{x}) \rangle \\ &\quad + \frac{1}{2} \|d\mathbf{x}\|_{\nabla_{\mathbf{x}}^2 \log p_{\mathcal{Y}|\mathcal{X}}(\mathbf{y}|\mathbf{x})}^2 + \mathcal{O}(\|d\mathbf{x}\|^3). \end{aligned}$$

Plugging in the Taylor expansion and using the identities $\int \pi(x) \nabla \log \pi(x) = 0$ and $\mathbf{G}(\mathbf{x}) = \mathbb{E}_{\mathbf{y}|\mathbf{x}} [-\nabla_{\mathbf{x}}^2 \log p_{\mathcal{Y}|\mathcal{X}}(\mathbf{y} | \mathbf{x})]$, we get

$$\text{KL}(p_{\mathcal{Y}|\mathcal{X}}(\mathbf{y} | \mathbf{x}) \| p_{\mathcal{Y}|\mathcal{X}}(\mathbf{y} | \mathbf{x} + d\mathbf{x})) = \frac{1}{2} d\mathbf{x}^{\top} \mathbf{G}(\mathbf{x}) d\mathbf{x} + \mathcal{O}(\|d\mathbf{x}\|^3).$$

Hence, the Fisher metric captures the local geometry of the likelihood under the KL-divergence. A remarkable property of the Fisher information is given by Čencov’s theorem [Čencov, 1982]. It states that the Fisher information metric is the unique Riemannian metric up to a rescaling constant that is invariant under sufficient statistical mappings.

Some downsides of the Fisher metric are computational. It depends on the probabilistic model and the expectation with respect to $\mathcal{Y} \mid \mathcal{X}$ might be intractable for some models, for example, if the likelihood is a Gaussian mixture. Another difficulty is that the metric is usually full-rank, meaning computation of the inverse $\mathbf{G}^{-1}(\mathbf{x})$ and the determinant have cubic complexity on D . If the exponential map computation requires numerical integration, then each integration step has cubic complexity because the Christoffel symbols (Eq. 2.3) use the metrics inverse.

The Fisher information metric (3.2) is defined in terms of the likelihood. However, in Bayesian inference one is interested in the posterior distribution, which depends on both $p_{\mathcal{Y}|\mathcal{X}}(\mathbf{y} \mid \mathbf{x})$ and the prior $p_{\mathcal{X}}(\mathbf{x})$. In this case, the Fisher information incorporates both terms, by adding the Hessian of the log-prior [Girolami and Calderhead, 2011],

$$\mathbf{G}(\mathbf{x}) = \mathbb{E}_{\mathcal{Y}|\mathcal{X}} \left[\nabla_{\mathbf{x}} \log p_{\mathcal{Y}|\mathcal{X}}(\mathbf{y} \mid \mathbf{x}) \nabla_{\mathbf{x}} \log p_{\mathcal{Y}|\mathcal{X}}(\mathbf{y} \mid \mathbf{x})^{\top} \right] + \nabla^2 \log p_{\mathcal{X}}(\mathbf{x}). \quad (3.3)$$

Monge metric Recall $\ell(\mathbf{x}) = \log p(\mathbf{x})$ and consider the graph of the tempered log-density

$$\mathcal{G}_{\alpha} := \{(\mathbf{x}, \alpha \ell(\mathbf{x})) : \mathbf{x} \in \mathbb{R}^D\} \subset \mathbb{R}^{D+1}, \quad \alpha \geq 0.$$

This is a D dimensional manifold embedded in $\mathbb{R}^{D+1}$ (See Sec. 2.1). The Monge metric is the Riemannian tensor on $\mathbb{R}^{D+1}$ given by this embedding [Hartmann et al., 2022],

$$\mathbf{G}(\mathbf{x}) = \mathbf{I}_D + \alpha^2 \nabla \ell(\mathbf{x}) \nabla \ell(\mathbf{x})^{\top}.$$

This metric is more efficient than the Fisher metric because the inverse, determinant and Christoffel symbols are computable and depend on first and second order derivatives of the log-density which are computed efficiently with automatic differentiation (hessian vector products) and enables the metric to be used in higher dimensional problems such as Bayesian neural networks [Yu et al., 2023, Bergamin et al., 2024]

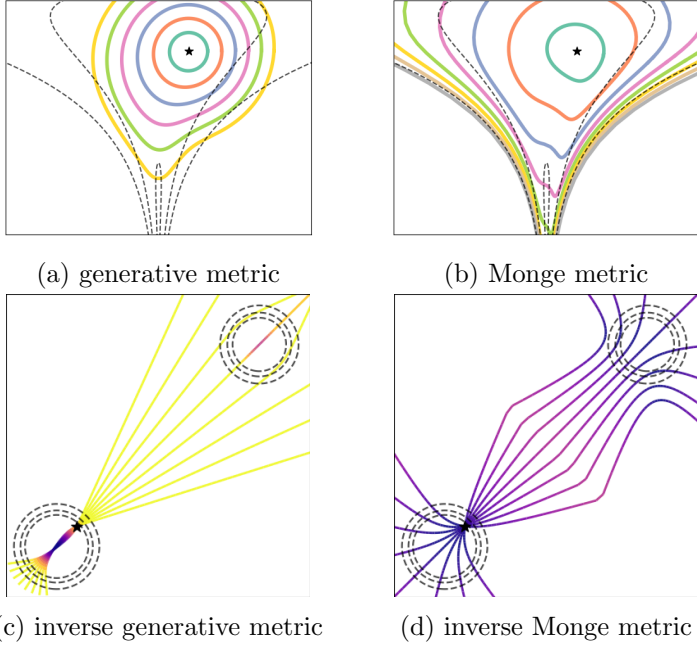


Figure 3.1: Visualization of the generative, inverse generative, Monge and inverse Monge metrics. **Top:** geodesic balls of varying radii centered at $\star$. **Bottom:** geodesics originating at $\star$ with different initial velocities (yellow indicates higher speed). The Monge metric deviates more from the Euclidean geometry than the generative metric, while the generative metric accelerates more in low-density regions.

The inverse metric and square-root metric are obtained by the Sherman-Morrison lemma

$$\begin{aligned}\mathbf{G}^{-1}(\mathbf{x}) &= \mathbf{I}_D - \frac{\alpha^2}{1 + \alpha^2 \|\nabla \ell(\mathbf{x})\|^2} \nabla \ell(\mathbf{x}) \nabla \ell(\mathbf{x})^\top, \\ \mathbf{G}^{1/2}(\mathbf{x}) &= \mathbf{I}_D + \frac{\alpha^2}{\sqrt{1 + \alpha^2 \|\nabla \ell(\mathbf{x})\|^2} + 1} \nabla \ell(\mathbf{x}) \nabla \ell(\mathbf{x})^\top.\end{aligned}$$

The determinant is $\det \mathbf{G}(\mathbf{x}) = 1 + \alpha^2 \|\nabla \ell(\mathbf{x})\|^2$ and the Christoffel symbols stacked into D matrices, of shape $D \times D$ are

$$\Gamma^k(\mathbf{x}) = \frac{\alpha^2 \partial_k \ell(\mathbf{x})}{1 + \alpha^2 \|\nabla \ell(\mathbf{x})\|^2} \nabla^2 \ell(\mathbf{x}). \quad (3.4)$$

The Monge metric has the property that it *slows down* geodesics that start from a local mode: the Euclidean distance covered by a geodesic is strictly smaller than its the distance covered by a straight line. Observation 1 formally states this phenomenon. This contraction is the factor causing the bias in Riemannian Laplace approximation under the Monge metric (See Section 3.2).

Observation 1. *Let $\mathbf{x}_0$ be the MAP estimate. Let $\mathbf{v}_0 \neq \mathbf{0}$ and let $\mathbf{x}_1 = \text{Exp}_{\mathbf{x}_0}(\mathbf{v}_0)$ denote the exponential map at time $t = 1$ under the Monge metric. Then*

$$\|\mathbf{x}_1 - \mathbf{x}_0\|_2 < \|\mathbf{v}_0\|_2.$$

Proof. Since $\nabla \ell(\mathbf{x}_0) = \mathbf{0}$, the Monge metric at $\mathbf{x}_0$ reduces to the identity, and the initial speed satisfies $\|\mathbf{v}_0\|_{\mathbf{G}(\mathbf{x})}^2 = \|\mathbf{v}_0\|_2^2$. Along any geodesic the metric speed is conserved, so

$$\|\mathbf{v}_0\|_2^2 = \|\dot{\mathbf{x}}(t)\|_{\mathbf{G}(\mathbf{x}(t))}^2 = \|\dot{\mathbf{x}}(t)\|_2^2 + \alpha^2 \langle \nabla \ell(\mathbf{x}(t)), \dot{\mathbf{x}}(t) \rangle^2, \quad \forall t \in [0, 1].$$

and rearranging gives

$$\|\dot{\mathbf{x}}(t)\|_2^2 = \|\mathbf{v}_0\|_2^2 - \alpha^2 \left(\frac{d}{dt} \ell(\mathbf{x}(t)) \right)^2 \leq \|\mathbf{v}_0\|_2^2,$$

with equality only when $\frac{d}{dt} \ell(\mathbf{x}(t)) = 0$. Differentiating $\frac{d}{dt} \ell(\mathbf{x}(t)) = \nabla \ell(\mathbf{x}(t))^\top \dot{\mathbf{x}}(t)$ at $t = 0$ and using $\nabla \ell(\mathbf{x}_0) = \mathbf{0}$ together with the fact that the geodesic acceleration vanishes at the mode, one obtains

$$\left. \frac{d}{dt} \left[\nabla \ell(\mathbf{x}(t))^\top \dot{\mathbf{x}}(t) \right] \right|_{t=0} = \mathbf{v}_0^\top \nabla^2 \ell(\mathbf{x}_0) \mathbf{v}_0.$$

Since $\nabla^2 \ell(\mathbf{x}_0) \prec 0$ and $\mathbf{v}_0 \neq \mathbf{0}$, this derivative is strictly negative. By continuity, there exists $\epsilon > 0$ such that

$$\frac{d}{dt} \ell(\mathbf{x}(t)) \neq 0 \implies \|\dot{\mathbf{x}}(t)\|_2 < \|\mathbf{v}_0\|_2, \quad \forall t \in (0, \epsilon).$$

By the triangle inequality

$$\|\mathbf{x}_1 - \mathbf{x}_0\|_2 = \left\| \int_0^1 \dot{\mathbf{x}}(t) dt \right\|_2 \leq \int_0^1 \|\dot{\mathbf{x}}(t)\|_2 dt < \int_0^1 \|\mathbf{v}_0\|_2 dt = \|\mathbf{v}_0\|_2,$$

where the strict inequality holds because $\|\dot{\mathbf{x}}(t)\|_2 \leq \|\mathbf{v}_0\|_2$ everywhere with strict inequality on $(0, \epsilon)$, a set of positive measure. $\square$

Generative metric The generative metric [Kim et al., 2024] rescales the Euclidean geometry according to the inverse target density. This metric makes motion through high density regions cheaper than motion through low density regions. As a result, geodesic curves tend to “prefer” areas where the target density is large, see Observation 2. Figure 3.1 illustrates the Monge and generative metrics capturing the geometry of a curved distribution. Define the generative metric

$$\mathbf{G}(\mathbf{x}) = \rho(\mathbf{x})^2 \mathbf{I}_D, \quad \rho(\mathbf{x}) := \frac{p_0 + \lambda}{p(\mathbf{x}) + \lambda},$$

where $p_0, \lambda > 0$ prevent the metric from becoming singular when $p(\mathbf{x})$ is small. Since this metric is the identity matrix scaled, the inverse and determinant are straight to compute. The Christoffel symbols depend only on the gradient of the scalar factor,

$$\Gamma^k(\mathbf{x}) = \nabla \log \rho(\mathbf{x}) \mathbf{e}_k^\top + \mathbf{e}_k \nabla \log \rho(\mathbf{x})^\top - \partial_k \log \rho(\mathbf{x}) \mathbf{I}_D.$$

Observation 2 (High-density regions are cheaper under the generative metric). *Let*

$$\mathbf{G}(\mathbf{x}) = \rho(\mathbf{x})^2 \mathbf{I}_D, \quad \rho(\mathbf{x}) = \frac{p_0 + \lambda}{p(\mathbf{x}) + \lambda}, \quad p_0, \lambda > 0.$$

Then the Riemannian length of a piecewise C^1 curve $\gamma : [0, 1] \rightarrow \mathbb{R}^D$ is

$$L_{\mathbf{G}}(\gamma) = \int_0^1 \|\dot{\gamma}(t)\|_{\mathbf{G}(\gamma(t))} dt = \int_0^1 \left(\frac{p_0 + \lambda}{p(\mathbf{x}) + \lambda} \right) \|\dot{\gamma}(t)\|_2 dt.$$

This weighted average results in geodesics that have a tradeoff between higher density values and shorter velocity Euclidean norms.

The inverse metrics The metric choices we have presented until now (Fisher, Monge and generative) were proposed by others. We propose using the inverse metric to sample multimodal distributions. In general, any Riemannian tensor that is positive definite and full rank is mathematically valid. In particular, the inverse of the metrics is also positive definite and defines a valid metric tensor. This motivates defining the inverse metric:

$$\boxed{\mathbf{G}_{\text{inv}}(\mathbf{x}) = \mathbf{G}^{-1}(\mathbf{x})} \quad (3.5)$$

The determinant is $\det \mathbf{G}_{\text{inv}}(\mathbf{x}) = 1/\det \mathbf{G}(\mathbf{x})$. The Christoffel symbols are analytic and computable with automatic differentiation for the Monge and generative metrics.

By Observations 1 and 2 the Monge and Generative metrics induce geodesics that tend to stay within regions of high density. By using the inverse metric we hope to “revert” this property. In regions of low density, both the generative and Monge metrics assign larger Euclidean metric values, so their inverses effectively “speed up” motion. As a result, geodesics can cover larger Euclidean distances over the same amount of geodesic time. This behavior is formalized in Observations 4 and 3. Figure 3.1 illustrates the properties of the inverse generative and inverse Monge metrics on a bimodal distribution.

Although this does not guarantee that a geodesic will move from one mode to another in dimensions $D \geq 2$, we empirically observe in Section 3.3 that the geodesics under the inverse Monge metric are capable of traversing between modes of multimodal distributions.

Observation 3 (Paper II). Let $p(\mathbf{x})$ be a smooth density function. Let $(\mathbf{x}_t, \mathbf{v}_t)$ be the geodesic flow with initial conditions $(\mathbf{x}_0, \mathbf{v}_0)$ with respect to the inverse Monge metric, such that $\mathbf{x}_0$ is a local maximum. Then $\|\mathbf{v}_t\|_2 \geq \|\mathbf{v}_0\|_2$ for all $t \neq 0$.

Observation 4 (Paper II). Let $p(\mathbf{x})$ be a smooth density function. Let $(\mathbf{x}_t, \mathbf{v}_t)$ be the geodesic flow with initial conditions $(\mathbf{x}_0, \mathbf{v}_0)$ such that $p(\mathbf{x}_0) > 0$ with respect to the inverse generative metric. Then, for t such that $p(\mathbf{x}_t) \rightarrow 0$ we have $\|\mathbf{v}_t\|_2 > \|\mathbf{v}_0\|_2$.

3.1.2 Volume form and density on the manifold

Geodesic balls In both methods considered later, we need to sample tangent vectors at a unit geodesic distance of a fixed point. For $\mathbf{x} \in \mathbb{R}^D$

the sphere of radius r is

$$\mathbb{S}_{\mathbf{G},r}^{D-1}(\mathbf{x}) := \{\mathbf{v} \in T_{\mathbf{x}}\mathbb{R}^D : \|\mathbf{v}\|_{\mathbf{G}(\mathbf{x})} = r\}.$$

In practice we need to sample uniformly from the unit sphere which can be done as follows:

1. Sample $\mathbf{v} \sim \mathcal{N}(\mathbf{0}, \mathbf{G}^{-1}(\mathbf{x}))$
2. Normalize $\mathbf{v} \leftarrow \mathbf{v} / \|\mathbf{v}\|_{\mathbf{G}(\mathbf{x})}$.

Note that sampling $\mathbf{v} \sim \mathcal{N}(\mathbf{0}, \mathbf{G}^{-1}(\mathbf{x}))$ requires computing the inverse square root matrix $\mathbf{G}^{-1/2}$ which requires generally Cholesky decomposition, but for the Monge, generative, inverse Monge and inverse generative metrics it admits an analytic form.

Hausdorff density The original target density $p(\mathbf{x})$ is defined with respect to Lebesgue measure in $\mathbb{R}^D$. Once a Riemannian metric is introduced, the natural reference measure becomes the Riemannian volume element [Lee, 2018], which is

$$d\text{vol}_{\mathbf{G}}(\mathbf{x}) = \sqrt{\det \mathbf{G}(\mathbf{x})} d\mathbf{x}.$$

If we want to represent the density with respect to the volume element instead of Lebesgue measure, it must be rescaled by the corresponding factor. We refer to this density as the Hausdorff density and write

$$p_{\mathcal{H}}(\mathbf{x}) := \frac{p(\mathbf{x})}{\sqrt{\det \mathbf{G}(\mathbf{x})}}. \quad (3.6)$$

This is consistent with change of variables formula. Consider the diffeomorphism $\mathbf{z} = \varphi(\mathbf{x})$, change of variables gives

$$p_{\mathcal{Z}}(\mathbf{z}) = p(\varphi^{-1}(\mathbf{z})) |\det \mathbf{J}_{\varphi^{-1}}(\mathbf{x})|.$$

If $\mathbf{G}$ is the push-forward of the Euclidean metric under φ , then $\mathbf{G}(\mathbf{z}) = \mathbf{J}_{\varphi^{-1}}(\mathbf{z})^{\top} \mathbf{J}_{\varphi^{-1}}(\mathbf{z})$, and hence $\sqrt{\det \mathbf{G}(\mathbf{z})} = |\det \mathbf{J}_{\varphi^{-1}}(\mathbf{z})|$, and the change of variables formula can be written as

$$p_{\mathcal{Z}}(\mathbf{z}) = p(\varphi^{-1}(\mathbf{z})) \sqrt{\det \mathbf{G}(\mathbf{z})}.$$

The density with respect to the Riemannian volume measure is

$$p_{\mathcal{H}}(\mathbf{z}) = \frac{p_{\mathcal{Z}}(\mathbf{z})}{\sqrt{\det \mathbf{G}(\mathbf{z})}}.$$

That is, the Hausdorff density removes the Jacobian volume factor and gets the density expressed with respect to the geometry induced by the metric.

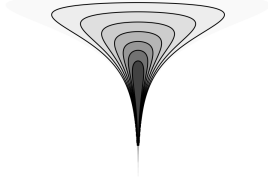
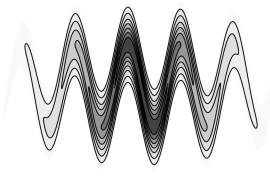
	Funnel	Rosenbrock	Squiggle
$\varphi(\mathbf{z})$	$\begin{bmatrix} e^{\sigma z_D/2} \mathbf{z}_{1:D-1} \\ \sigma z_D \end{bmatrix}$	$\begin{bmatrix} a + \frac{1}{\sqrt{2}} z_1 \\ \left(a + \frac{1}{\sqrt{2}} z_1\right)^2 + \frac{1}{\sqrt{2b}} z_2 \end{bmatrix}$	$\begin{bmatrix} z_1 \\ \mathbf{z}_{2:D} - \sin(az_1) \end{bmatrix}$
$p(\mathbf{x})$	$\mathcal{N}(x_D \mid 0, \sigma^2)$ $\times \mathcal{N}(\mathbf{x}_{1:D-1} \mid \mathbf{0}, e^{x_D} \mathbf{I}_{D-1})$	$\mathcal{N}(x_1 \mid a, \frac{1}{2})$ $\times \mathcal{N}(x_2 \mid x_1^2, \frac{1}{2b})$	$\mathcal{N}(\varphi^{-1}(\mathbf{x}) \mid \boldsymbol{\mu}, \boldsymbol{\Sigma})$ $\times \det \mathbf{J}_{\varphi^{-1}}(\mathbf{x}) $
			

Table 3.1: Three benchmark distributions with strong curvature used in the experiments.

3.1.3 Numerical computation of geodesics

For the Euclidean metric we can easily compute the exponential and logarithmic maps as in Eqs. 3.1, but for the other metric choices we solve the geodesic equations numerically. Let $\mathbf{x}(t)$ denote the geodesic trajectory and $\mathbf{v}(t) = \dot{\mathbf{x}}(t)$ velocity respectively, Equation 2.4 is rewritten as

$$\begin{aligned} \dot{\mathbf{x}}(t) &= \mathbf{v}(t), \\ \dot{\mathbf{v}}_k(t) &= -\mathbf{v}(t)^\top \Gamma^k(\mathbf{x}(t)) \mathbf{v}(t), \quad k = 1, \dots, D. \end{aligned} \quad (3.7)$$

where $\Gamma^k(\mathbf{x}) \in \mathbb{R}^{D \times D}$ is the matrix with the Christoffel symbols (Eq. 2.3). The main computational bottleneck comes from the Christoffel symbols, since they involve the inverse metric and derivatives of the metric. The Fisher metric in general has $\mathcal{O}(D^3)$ complexity, the Monge and inverse Monge metrics have the complexity of Hessian-vector products, and the generative and inverse generative metrics have the complexity of gradient computations.

The choice of numerical solver affects accuracy and runtime [Kidger, 2022]. Fixed step-size, adaptive and higher order solvers introduce trade-offs between stability, numerical error and computational cost. We conduct empirical studies on the choice of numerical solver for different metric choices in Section 3.3.

3.1.4 Target distributions with strong curvature

The need for a non-Euclidean metric is most pronounced in regions of the parameter space that are narrow along certain directions but still carry significant probability mass. For evaluation of these methods we hence consider three distributions with strong curvature that are constructed from a diffeomorphism of a standard Gaussian. They are the Funnel, Hybrid Rosenbrock and Squiggle distributions. Here we present a simpler version of the Hybrid Rosenbrock, the D -dimensional is given in Pagani et al. [2022]. Table 3.1 shoes the distributions, summarizes the transformations and the corresponding target densities.

The latent value is distributed $\mathbf{z} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ and $\mathbf{x} = \varphi(\mathbf{z})$. The bijection φ allows the generation of samples for benchmarking. The pullback metric follows from the transformation rule of Riemannian tensors,

$$\mathbf{G}(\mathbf{x}) = \mathbf{J}_{\varphi^{-1}}(\mathbf{x})^\top \boldsymbol{\Sigma}^{-1} \mathbf{J}_{\varphi^{-1}}(\mathbf{x}). \quad (3.8)$$

3.2 Case study I: Riemannian Laplace approximation

We apply the geometric tools we have described above in the construction of Riemannian Laplace approximation (RLA) [Bergamin et al., 2024]. The key idea is to replace straight lines in Euclidean Laplace approximation with geodesics. A Gaussian distribution is formed on the tangent space at the mode. Then the velocities are pushed forward to the parameter space with the exponential map, in this way the approximation adapts to the geometry (given by the metric tensor) of the target distribution (see Fig. 3.2).

Let f be a scalar output function defined on the manifold and $c(t)$ a curve on the manifold such that $c(0) = \mathbf{x}$, the second order Taylor approximation is [Boumal, 2023][Section 5.9, Eq. 5.26]

$$\begin{aligned} f(c(t)) &= f(\mathbf{x}) + t \langle \nabla_{\mathbf{G}} f(\mathbf{x}), \mathbf{v} \rangle_{\mathbf{G}(\mathbf{x})} + \frac{t^2}{2} \langle \text{Hess}_{\mathbf{G}} f(\mathbf{x})[\mathbf{v}], \mathbf{v} \rangle_{\mathbf{G}(\mathbf{x})} \\ &\quad + \frac{t^2}{2} \langle \nabla_{\mathbf{G}} f(\mathbf{x}), c''(0) \rangle_{\mathbf{G}(\mathbf{x})} + \mathcal{O}(t^3). \end{aligned} \quad (3.9)$$

Let $\mathbf{x}$ be a local maximum of the target logdensity $\ell(\mathbf{x})$. In the case of a probabilistic model $\mathbf{x}$ is the maximum a posterior (MAP) estimate. At the mode both Euclidean and Riemannian gradients vanish, and a second order Taylor expansion of ℓ along geodesics (Eq. 3.9) simplifies to

$$\ell(\text{Exp}_{\hat{\mathbf{x}}}(\mathbf{v})) \approx \ell(\hat{\mathbf{x}}) + \frac{1}{2} \langle \text{Hess}_{\mathbf{G}} \ell(\hat{\mathbf{x}})[\mathbf{v}], \mathbf{v} \rangle_{\mathbf{G}(\hat{\mathbf{x}})}, \quad (3.10)$$

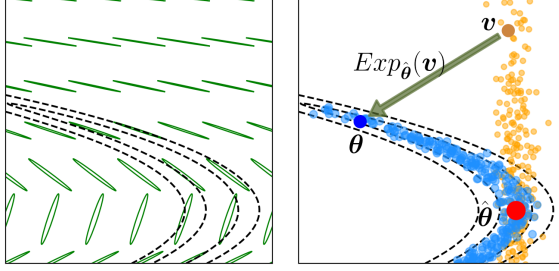


Figure 3.2: Visualization of Riemannian Laplace approximation. The Fisher metric (green) captures the local curvature. Right: Samples from a Gaussian distribution (orange) are deterministically mapped using the Fisher metric to provide a flexible approximation (blue).

Algorithm 3 RLA, producing N samples $\mathbf{x}^{(n)}$.

Require: Metric $\mathbf{G}$ and covariance Σ

- 1: Obtain the MAP estimate $\hat{\mathbf{x}}$.
 - 2: **for** $n \leftarrow 1 \dots N$ **do**
 - 3: Obtain $\mathbf{v}^{(n)} \sim \mathcal{N}(\mathbf{0}, \Sigma)$
 - 4: Push-forward $\mathbf{x}^{(n)} \leftarrow \text{Exp}_{\hat{\mathbf{x}}}(\mathbf{v}^{(n)})$
-

where t is absorbed into $\mathbf{v} \leftarrow t\mathbf{v}$. For the density itself,

$$p(\text{Exp}_{\hat{\mathbf{x}}}(\mathbf{v})) \approx p(\hat{\mathbf{x}}) \exp\left\{\frac{1}{2} \langle \text{Hess}_{\mathbf{G}} \ell(\hat{\mathbf{x}})[\mathbf{v}], \mathbf{v} \rangle_{\mathbf{G}(\hat{\mathbf{x}})}\right\}. \quad (3.11)$$

At the mode the Riemannian Hessian components (Eq. 2.6) simplify to $\text{Hess}_{\mathbf{G}} \ell(\hat{\mathbf{x}})[\mathbf{v}] = \mathbf{G}^{-1}(\hat{\mathbf{x}}) \nabla^2 \ell(\hat{\mathbf{x}}) \mathbf{v}$, since $\nabla \ell(\hat{\mathbf{x}}) = \mathbf{0}$ (personal communication with Marcelo Hartmann). Note that $\mathbf{G}^{-1}$ and $\mathbf{G}$ multiply, thus, the quadratic term is the Euclidean Hessian of ℓ , which motivates a Gaussian approximation in the tangent space at $\hat{\mathbf{x}}$ with $\mathcal{N}(\mathbf{0}, \Sigma^{-1})$, where $\Sigma = -\nabla^2 \ell(\hat{\mathbf{x}})$. Samples that approximate the target are generated by mapping the exponential map on the tangent velocities, see Algorithm 3.

Bergamin et al. [2024] proposed using the Monge metric in the RLA approximation, which we denote RLA-B. Although the Monge metric has lower complexity than Fisher metric, this approximation is biased for Gaussian targets as a consequence of Observation 1. The Monge geodesics contract Euclidean distances near a mode and samples generated by the geodesics under this metric are too close to the mode, while the Euclidean Laplace is exact.

Paper I proposes two improvements that correct this behavior:

- Apply a logarithm map before to correct the magnitude of the velocities. Namely, replace step 3 in Algorithm 3 with

$$\begin{aligned}\tilde{\mathbf{x}}^{(n)} &\sim \mathcal{N}(\hat{\mathbf{x}}, (-\nabla^2 \ell(\hat{\mathbf{x}}))^{-1}), \\ \mathbf{v}^{(n)} &= \text{Log}_{\mathcal{P}, \hat{\mathbf{x}}}(\tilde{\mathbf{x}}^{(n)}),\end{aligned}$$

where $\text{Log}_{\mathcal{P}, \hat{\mathbf{x}}}$ is computed with respect to the Monge metric induced by the density $\mathcal{N}(\hat{\mathbf{x}}, (-\nabla^2 \ell(\hat{\mathbf{x}}))^{-1})$, we refer to this variant as RLA-BLog.

- Use the Fisher information metric in Riemannian Laplace approximation, denoted RLA-F.

Additionally, Paper I studies choices that matter in practice: whether the approximation should be centered at the Euclidean MAP or at the Hausdorff MAP and whether the $\Sigma = (-\nabla^2 \ell(\hat{\mathbf{x}}))^{-1}$ or $\Sigma = \mathbf{G}^{-1}(\hat{\mathbf{x}})$. The Fisher information matrix has theoretical advantages: it is exact for Gaussian targets, asymptotically for targets that are Gaussian on the limit of infinite data, and it is exact for targets that are diffeomorphic to a Gaussian. Theorem 3.1 states the exactness of RLA-F for Gaussian diffeomorphism.

Theorem 3.1 (Paper I - Theorem 3). *With an invariant prior, e.g. Jeffreys prior, RLA-F with Hausdorff MAP is exact for probabilistic models whose target distributions are diffeomorphic with Gaussians, for which the negative Hessian at the Hausdorff MAP coincides with the Fisher metric.*

3.2.1 Empirical evaluation

The empirical study compares Euclidean Laplace approximation (ELA), the Monge-based approximation (RLA-B), the logarithm-map corrected Monge variant (RLA-BLog), and the Fisher variant (RLA-F). Here we include two of the experiments from Paper I, the objective of the first experiment is to visually illustrate each of the methods and the effect of the Hausdorff MAP. The second experiment shows that even though RLA-F is more expensive than Monge due to the cubic complexity, it can be more efficient in the number of steps needed by an adaptive numerical integrator which makes the RLA-F the recommended choice for problems where the Fisher is computable and with tens to low hundred of dimensions. Since all Riemannian variants require solving geodesic equations numerically, in all experiments we use the same Dopri5 adaptive integrator [Kidger, 2022].

		ELA	RLA-B		RLA-BLog		RLA-F	
	data	$\mathcal{W}$	$\mathcal{W} \downarrow$	T	$\mathcal{W} \downarrow$	T	$\mathcal{W} \downarrow$	T
stand.	Ripl	0.106 ± 0.004	0.236 ± 0.001	29.7	0.16 ± 0.065	45.2	0.064 ± 0.002	12.2
	Pima	0.149 ± 0.0	0.274 ± 0.0	41.7	0.21 ± 0.007	78.5	0.147 ± 0.0	12.2
	Hear	0.529 ± 0.001	0.649 ± 0.0	40.2	1.154 ± 0.189	71.6	0.514 ± 0.0	13.2
	Aust	0.441 ± 0.001	0.522 ± 0.001	51.5	0.745 ± 0.203	79.1	0.417 ± 0.001	18.0
	Germ	0.388 ± 0.001	0.431 ± 0.0	58.4	0.676 ± 0.1	104.0	0.387 ± 0.0	14.2
raw	Ripl	0.437 ± 0.017	0.489 ± 0.012	30.2	0.554 ± 0.253	45.0	0.247 ± 0.012	12.7
	Pima	0.21 ± 0.01	0.294 ± 0.004	5632.9	0.216 ± 0.014	1690.4	0.112 ± 0.008	15.4
	Hear	0.842 ± 0.026	0.96 ± 0.012	<i>7868.2</i>	0.898 ± 0.029	<i>3161.1</i>	0.644 ± 0.012	17.9
	Aust	0.454 ± 0.004	0.455 ± 0.01	<i>18513.8</i>	0.467 ± 0.005	<i>12298.6</i>	0.378 ± 0.005	18.0
	Germ	0.846 ± 0.002	0.92 ± 0.003	3545.9	0.898 ± 0.003	2985.5	0.823 ± 0.001	17.8

Table 3.2: Logistic regression results as mean $\pm$ std. Bold font indicates the best method. Italic font indicates the integrator reached maximum number of steps in at least one run. $\mathcal{W}$ indicates Wasserstein distance to NUTS samples while T indicates the average number of function evaluations for one sample. For all evaluation metrics smaller is better.

Visual illustration of Riemannian Laplace The first experiment is a visual comparison on the Banana distribution. Figure 3.3 shows the approximations obtained from both the Euclidean target density (first column) and the corresponding Hausdorff density (second column). The take-away message is that Euclidean Laplace approximation does not capture the curvature, and among the geometry-aware methods the Fisher information produces the best fit to the target distribution, note how RLA-B samples do not properly spread out without the Log-corrector.

Efficiency of the approximation The second experiment considers Bayesian logistic regression on several benchmark data sets, both with standardized inputs and with raw inputs. The Fisher metric includes the Hessian of the prior as in Eq. 3.3. Table 3.2 reports the Wasserstein distance to reference samples obtained with NUTS and the number of average number of log-density evaluations per sample. The main takeaway is that RLA-F is more efficient in integration steps (geodesics are easier to integrate) and is more accurate than the Euclidean and Monge metric variants.

3.3 Case study II: geodesic slice sampling

This case study uses the geometric tools from Section 3.1 in an MCMC sampling method. In Section 2.4.1 we introduced hit-and-run slice sampling that does slice sampling along Euclidean straight lines, geodesic slice sampling from Paper II keeps the same logic but replaces the straight lines

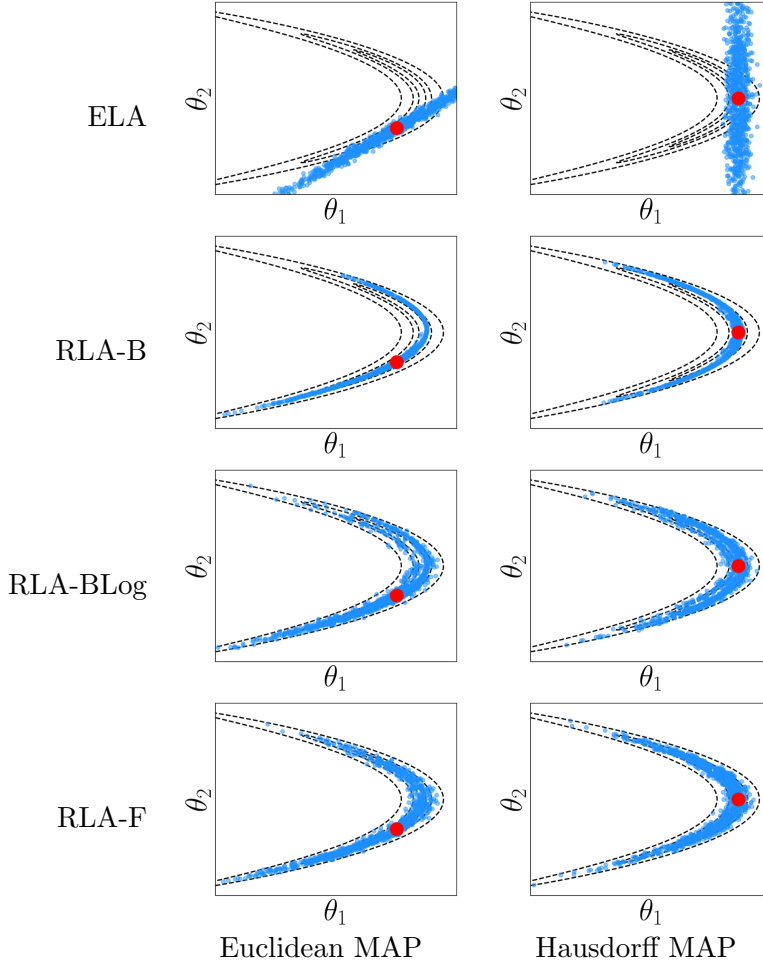


Figure 3.3: Comparison of Euclidean and Hausdorff MAP estimates. The red points are the Euclidean or Hausdorff MAP, and samples are scatter points in blue. The Euclidean metric (ELA) does not capture the geometry. The Monge metric is narrow without the logarithm map adjustment (RLA-B). The Fisher metric(RLA-F) is the most accurate with the Hausdorff MAP.

by the geodesics induced by a metric choice. This change allows for MCMC sampling of distributions with strong curvature, multimodality or both.

Geodesic slice sampling Our method builds on Durmus et al. [2026], that introduced geodesic slice sampling over manifolds with closed form geodesics (e.g the sphere, torus, hyperbolic space,...). Paper II extends this method to distributions defined on $\mathbb{R}^D$ with generic metric tensors $\mathbf{G}(\mathbf{x})$. Our method is metric-agnostic geodesic slice sampling (MAGSS).

Let $\mathbf{x}^{(n)}$ denote the current state. In Euclidean hit-and-run slice sampling one first samples a random velocity $\mathbf{v}^{(n)}$ and performs univariate slice sampling on the straight line passing through $\mathbf{x}^{(n)}$. Our method replaces the straight lines with geodesics, the density with the Hausdorff density and the Euclidean sphere with the Riemannian sphere (See Algorithm 4) Denote by $\hat{\gamma}_{\mathbf{x},\mathbf{v}}(t)$ the numerical solution of geodesic equations with initial conditions $\mathbf{x}, \mathbf{v}$ for time t . Table 3.3 shows the changes between Euclidean hit-and-run slice sampling and MAGSS.

Choice of metric Similar to Riemannian Laplace, the main modeling decision is the choice of Riemannian metric. For targets with strong local curvature, the Fisher, Monge and generative metrics can make geodesics better aligned with the shape of the density, with tradeoffs in the computational speed.

For multimodal targets, the inverse Monge and inverse generative metrics play a different role: they increase Euclidean speed in low-density regions and might be able to bridge between different modes, as discussed in Observations 3 and 4. In this sense, the same sampler can be adapted either to local geometry or to global mode exploration by changing only the metric tensor (as shown in Fig. 3.1). The Monge and generative metrics depend on hyperparameters. A limitation of our method, is that there is no default value for these metric-parameters, some problems might allow for computation of a validation statistic for selection but no general value is optimal.

Paper II also considers a simple meta-sampler that alternates between geodesic slice sampling moves and local MCMC updates (e.g. MALA). The purpose of this combination is to use MAGSS with the inverse metrics for global transitions and the local Markov chain for exploration within each mode. This is particularly useful when the target is both multimodal and locally has high curvature. This is denoted by Meta-MAGSS.

	Euclidean	Riemannian
Slice level	$s \sim \text{Unif}(0, p(\mathbf{x}^{(n)}))$	$s \sim \text{Unif}(0, \textcolor{red}{p}_{\mathcal{H}}(\mathbf{x}^{(n)}))$
Velocity	$\mathbf{v}^{(n)} \sim \text{Unif}(\mathbb{S}^{D-1}(\mathbf{x}^{(n)}))$	$\mathbf{v}^{(n)} \sim \text{Unif}(\mathbb{S}_{\mathbf{G}}^{D-1}(\mathbf{x}^{(n)}))$
Slice sampling on	$\gamma_{\mathbf{x}^{(n)}, \mathbf{v}^{(n)}}(t) = \mathbf{x}^{(n)} + t\mathbf{v}^{(n)}$	$\textcolor{red}{\hat{\gamma}}_{\mathbf{x}^{(n)}, \mathbf{v}^{(n)}}(t) = \textcolor{red}{\text{Exp}}_{\mathbf{x}^{(n)}}(t\mathbf{v}^{(n)})$

Table 3.3: Comparison of Euclidean and Riemannian slice sampling components. Red highlights indicate the Riemannian modifications.

Algorithm 4 Geodesic slice sampling, producing N samples $\mathbf{x}^{(n)}$.

Require: Initial $\mathbf{x}^{(0)}$, metric $\mathbf{G}$ and parameters $m \in \mathbb{N}$, $w \geq 0$

- 1: **for** $n \leftarrow 0 \dots N - 1$ **do**
 - 2: Sample $s \sim \text{Unif}(0, p_{\mathcal{H}}(\mathbf{x}^{(n)}))$ and $\mathbf{v}^{(n)} \sim \text{Unif}(\mathbb{S}_{\mathbf{G}}^{D-1}(\mathbf{x}^{(n)}))$
 - 3: Obtain $(\ell, r) = \text{Step-out}_{w,m}(s, \hat{\gamma}_{\mathbf{x}^{(n)}, \mathbf{v}^{(n)}})$
 - 4: Sample $t^* = \text{Shrink}_{\ell,r}(s, \hat{\gamma}_{\mathbf{x}^{(n)}, \mathbf{v}^{(n)}})$
 - 5: Push-forward $\mathbf{x}^{(n+1)} \leftarrow \hat{\gamma}_{\mathbf{x}^{(n)}, \mathbf{v}^{(n)}}(t^*)$
-

3.3.1 Empirical evaluation

The empirical evaluation compares parallel tempering [Łatuszyński et al., 2025] (PT), diffusion Gibbs sampler [Chen and Lipman, 2024] (DiGS) and metric-agnostic geodesic slice sampling (MAGSS).

We highlight here some of the experiments from Paper II. The objective of the experiments is to study our method on curved distributions and multimodal distributions, comparing the accuracy and running time against the alternatives.

The first experiment shows MAGSS performance on curved target distributions. The second experiment compares the mode exploration of the methods on a highly multimodal target distribution modeling the Allen-Cahn field system. The third experiment studies the effect of the numerical integrator on our method. Refer to the paper for full description of the experiment setup and further analysis.

Curved target distributions We consider the target distributions with strong curvature from Section 3.1.4 (Funnel, Rosenbrock and Squiggle). The Fisher metric in the plots is the pullback metric of the transformation. Figure 3.4 shows the empirical Wasserstein distance to reference samples from the distributions for dimensions 2, 4, ..2⁸. The pullback metric produces the highest quality samples, but becomes computationally too expensive as can be seen by the cut-off in the lines, the Monge metric produces

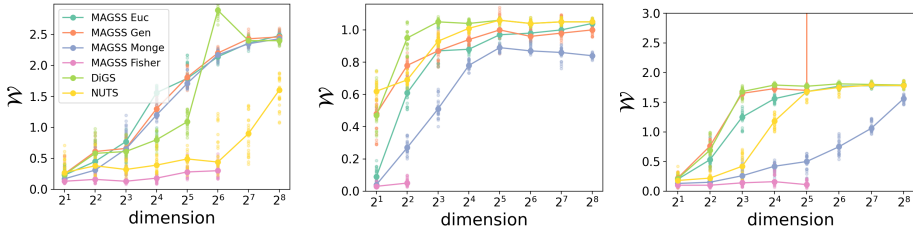


Figure 3.4: Univariate sampling accuracy on different choice of Riemannian metric (Wasserstein distance, lower is better) for targets of varying dimensionality. The medians over 5 runs are connected with a line. Left: Funnel. Middle: Rosenbrock. Right: Squiggle.

good results on Rosenbrock and Squiggle distributions. DiGS, PT and NUTS struggle exploring the curved regions of the space and are less accurate, note that DiGS use the local MCMC Metropolis-adjusted Langevin algorithm (MALA), a Riemannian version could be use instead to enhance exploration of the curved regions as an improvement to the method. The values of the hyperparameters of the Monge and generative metrics are $\alpha^2, \lambda = 1$.

Highly multimodal target We consider the Allen-Cahn field system which has 2^D modes with $D = 16$. This model has two global modes at $\mathbf{x} = \pm \mathbf{1}$, and multiple local modes at values $\mathbf{x} = (\pm 1, \dots, \pm 1)$. Table 3.4 shows the results of the different sampling method. Since there is no access to reference samples, we compute the Kernel Stein Discrepancy, but it is not trustworthy since it does not account for multimodality, the table also reports the percentage of samples such that $x_8 \geq 0$ to see if the method was stuck on a single mode and the running time in seconds. MAGSS and DiGS are one order of magnitude slower than PT since they both rely on numerical integration. Figure 3.5 shows that MAGSS explores the global and local modes, jumping the most frequently in between, while PT explores only the global modes and DiGS collapses to a single mode. A log-10 grid of parameters α^2, λ were tried, selecting the value based on the jump percentage.

Sensitivity to the numerical integrator Here we study the effect of the choice numerical integrator on a two dimensional 40-mode Gaussian multimodal target for different values of α^2 and λ on a log-10 scale. We test integrators that are adaptive or fixed step-size and integrators with

Table 3.4: Allen-Cahn system model

sampler	KSD V-stat	$x_8 > 0$	t(s)
PT	0.13 ± 0.04	0.35	6
DiGS	0.13 ± 0.05	0.0	157
Meta-MAGSS	2.98 ± 0.7	0.33	32

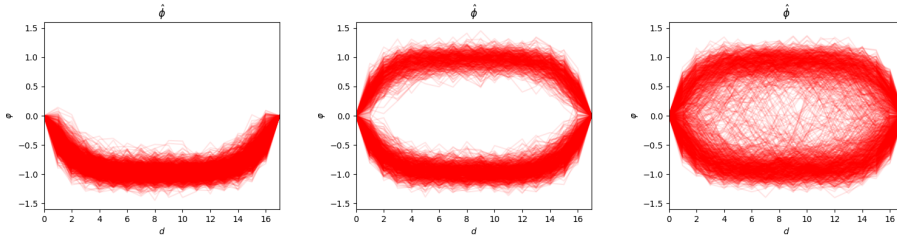


Figure 3.5: Samples from Allen-Cahn field system model with 2^{16} modes that zig-zag between -1 and 1 on the y-axis over the $D = 16$ values at the x-axis. Euclidean DiGS (left) gets stuck in one mode, parallel tempering (middle) only explores two dominant modes with constant value over the x-axis, whereas Meta-MAGSS (right; inverse generative with $\lambda = 10^{-6}$) explores also the modes that switch between the extremes.

different order accurate solutions (see Figure 3.6). The general trend as the inverse Monge and inverse generative diverge from the Euclidean (α^2 grows, λ reduces) the system is more stiff and takes longer to numerical integrate. For the inverse Monge metric higher order and adaptive integrators work better, in practice we used Dopri5 in the other experiments because it has a good balance between the two.

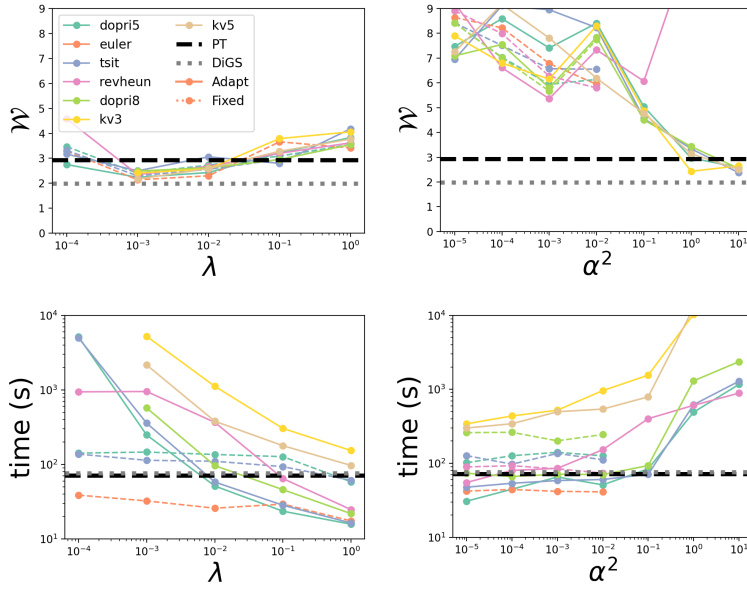


Figure 3.6: Different numerical integrators for the 40 mode Gaussian mixture model. The black dotted lines are PT and DiGS. Left: inverse generative metric with parameter λ . Right: inverse Monge metric with parameter α^2 .

Chapter 4

Geometric Modeling of Discrete Data

In this chapter we propose using the Aitchison geometry of the simplex space for modeling discrete data. The main idea is to represent simplex-valued distributions in Euclidean coordinates in a way that respects their natural geometry. We consider two applications of this idea: generative modeling of discrete data (Paper III) and Gaussian process classification (Paper IV).

We first introduce the Aitchison geometry and the simplex-to-Euclidean bijections that are used through the chapter. We then present stochastic interpolations that move discrete data from the boundary of the simplex to the interior, and provide theoretical guarantees for the recovery of the original discrete values. In the interior of the space we can use the bijections for modeling continuously in Euclidean coordinates, making the modeling compatible with Euclidean continuous methods. These ingredients are combined in two case studies: Simplex-to-Euclidean Flow Matching for generation of discrete data and Isometric Logratio Gaussian Processes for multiclass classification.

Motivation We find that the Aitchison geometry provides for the simplex, a geometry-aware representation that, combined with the stochastic interpolations, connects discrete modeling to known Euclidean methods. This makes existing Euclidean methods applicable to these tasks (generation and classification), providing a unified modeling framework that is exact and compatible with state-of-the-art Euclidean models while respecting the geometry of log proportions.

4.1 Simplex-to-Euclidean Bijections

4.1.1 Compositional Data Analysis

We presented in Section 2.5 the Fisher information geometry in the simplex, which is isomorphic to the positive orthant of the sphere. In this chapter we instead use the Aitchison geometry of the simplex, because it provides a direct connection between the open simplex and Euclidean coordinates.

Aitchison [1981, 1982] introduces a geometry for the simplex that is not the Fisher information metric. This geometry is defined only on the interior of the simplex and the central idea is that information is carried by the log-ratios of the components. Similarly to how the sigmoid function is a link function from binary classification to $\mathbb{R}$, the Aitchison geometry gives different choices of link functions from the open simplex to $\mathbb{R}^D$. The bijections make Euclidean modeling compatible with the simplex.

The geometry of the simplex Consider K -class categorical data $c \in \{1, \dots, K\}$, let $D := K - 1$. The simplex and the open simplex are defined as

$$\Delta^D := \{\mathbf{x} \in \mathbb{R}_{\geq 0}^K : \sum_{k=1}^K x_k = 1\}, \quad \mathring{\Delta}^D := \{\mathbf{x} \in \mathbb{R}_{> 0}^K : \sum_{k=1}^K x_k = 1\},$$

where only $D = K - 1$ components are free, since one arbitrary component (for example the last one) is determined by the others as $x_K = 1 - \sum_{d=1}^D x_d$. The usual Euclidean operators are not closed on the simplex, for example the Euclidean sum of two vectors can fall outside the space. For this reason, Aitchison [1986] defines perturbation (vector addition) and powering (scalar multiplication) that make the simplex a vector space. Define the normalization operation $\mathcal{C}(\mathbf{x}) := \mathbf{x} / \sum_{k=1}^K x_k$. For $\mathbf{x}, \mathbf{y} \in \mathring{\Delta}^D$ and $\lambda \in \mathbb{R}$, define

$$\begin{aligned} \text{Perturbation} \quad \mathbf{x} \oplus \mathbf{y} &:= \mathcal{C}(x_1 y_1, \dots, x_K y_K), \\ \text{Powering} \quad \lambda \odot \mathbf{x} &:= \mathcal{C}(x_1^\lambda, \dots, x_K^\lambda). \end{aligned}$$

Under these operations, $(\mathring{\Delta}^D, \oplus, \odot)$ is a vector space and the neutral element is $\frac{1}{K} \mathbf{1}$ since $\mathbf{x} \oplus (\frac{1}{K} \mathbf{1}) = \mathbf{x}$. The vector space is an inner product space when given the Aitchison inner product defined as

$$\langle \mathbf{x}, \mathbf{y} \rangle_A := \frac{1}{2K} \sum_{i,j=1}^K \log \frac{x_i}{x_j} \log \frac{y_i}{y_j}.$$

The norm of a vector is $\|\mathbf{x}\|_A = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle_A}$. The distance between two compositions is

$$d_A(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} \ominus \mathbf{y}\|_A = \sqrt{\frac{1}{2K} \sum_{i,j=1}^K \left(\log \frac{x_i}{x_j} - \log \frac{y_i}{y_j} \right)^2},$$

where $\mathbf{x} \ominus \mathbf{y} := \mathbf{x} \oplus (-1 \odot \mathbf{y})$.

The simplex as embedded Riemannian manifold The simplex can be viewed as an embedded manifold in $\mathbb{R}^K$. It is the intersection of the hyperplane $\sum_{k=1}^K x_k = 1$ and the positive orthant. Because the hyperplane is flat, all tangent spaces are parallel and identified with the zero-sum linear space

$$\mathbb{H}^D = \left\{ \mathbf{z} \in \mathbb{R}^K : \sum_{i=1}^K z_i = 0 \right\}.$$

To give the open simplex a geometry, we use the center logratio map (CLR), which is a bijection between $\mathring{\Delta}^D$ and the zero-sum hyperplane $\mathbb{H}^D$:

$$\begin{aligned} \text{clr} : \mathring{\Delta}^D &\rightarrow \mathbb{H}^D, & \mathbf{x} &\mapsto \mathbf{z} = \log \mathbf{x} - \frac{1}{K} \sum_{k=1}^K \log x_k \mathbf{1}, \\ \text{clr}^{-1} : \mathbb{H}^D &\rightarrow \mathring{\Delta}^D, & \mathbf{z} &\mapsto \mathbf{x} = \mathcal{C}(\exp \mathbf{z}). \end{aligned}$$

The image of the map lies in $\mathbb{H}^D$ because the components of $\text{clr}(\mathbf{x})$ sum to zero, and the inverse map is usually known as the *softmax*. In addition, the center logratio map is an isometry between $(\mathring{\Delta}^D, \langle \cdot, \cdot \rangle_A)$ and $(\mathbb{H}^D, \langle \cdot, \cdot \rangle_2)$, since the Aitchison norm and inner products are (after some algebraic manipulations):

$$\begin{aligned} \langle \mathbf{x}, \mathbf{y} \rangle_A &= \langle \text{clr}(\mathbf{x}), \text{clr}(\mathbf{y}) \rangle_2 = \frac{1}{2K} \sum_{i,j=1}^K \log \frac{x_i}{x_j} \log \frac{y_i}{y_j}, \\ d(x, y)_A &= \|\text{clr}(\mathbf{x}) - \text{clr}(\mathbf{y})\|_2 = \sqrt{\frac{1}{2K} \sum_{i,j=1}^K \left(\log \frac{x_i}{x_j} - \log \frac{y_i}{y_j} \right)^2}. \end{aligned}$$

This means that the Aitchison geometry of the simplex is inherited from $\mathbb{H}^D$ via the center logratio map, thus the Riemannian metric is the pullback metric:

$$\mathbf{G}(\mathbf{x}) = \mathbf{J}_{\text{clr}}(\mathbf{x})^\top \mathbf{J}_{\text{clr}}(\mathbf{x}),$$

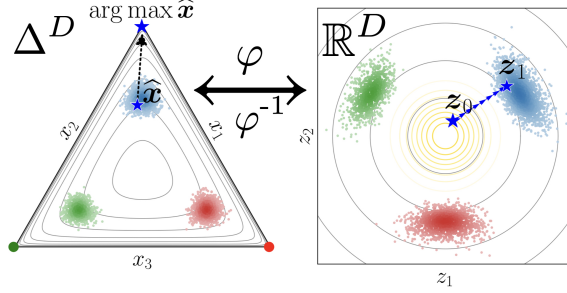


Figure 4.1: Visual representation of Simplex-to-Euclidean Flow Matching.

for $\mathbf{J}_{\text{clr}}(\mathbf{x}) = \frac{\partial \text{clr}(\mathbf{x})}{\partial \mathbf{x}_{1:D}}$ where the Jacobian is computed with respect to the first D coordinates because of the extra degree of freedom and $x_K = 1 - \sum_{d=1}^D x_d$. The geodesics are computed by pulling back straight lines in $\mathbb{H}^D$

$$\gamma(t) = \text{clr}^{-1}((1-t)\text{clr}(\mathbf{x}) + t\text{clr}(\mathbf{y})) = ((1-t) \odot \mathbf{x}) \oplus (t \odot \mathbf{y}).$$

Isometric Logratio The CLR map gives the simplex the Aitchison geometry, but the image of the map still lands in the constrained hyperplane $\mathbb{H}^D$. For practical modeling it is more convenient to work in ordinary Euclidean coordinates. The isometric logratio (ILR) map of Egozcue et al. [2003] achieves this by expressing CLR coordinates in an orthonormal basis of $\mathbb{H}^D$. This basis is given by the Helmert matrix $\mathbf{H} \in \mathbb{R}^{D \times K}$ [Lancaster, 1965], and the resulting bijection is

$$\varphi(\mathbf{x}) = \mathbf{H} \text{clr}(\mathbf{x}).$$

Since the Helmert matrix has the property that $\mathbf{H}\mathbf{1} = \mathbf{0}$, this is equivalently written as

$$\begin{aligned} \varphi : \mathring{\Delta}^D &\rightarrow \mathbb{R}^D, & \mathbf{x} &\mapsto \mathbf{z} = \mathbf{H} \log \mathbf{x}, \\ \varphi^{-1} : \mathbb{R}^D &\rightarrow \mathring{\Delta}^D, & \mathbf{z} &\mapsto \mathbf{x} = \mathcal{C} \left(\exp(\mathbf{H}^\top \mathbf{z}) \right). \end{aligned} \quad (4.1)$$

This bijection is simpler because we can use the standard basis of the Euclidean space $\mathbb{R}^D$ along with the Euclidean inner product to model in the latent space while preserving the Aitchison geometry. The isometry is stated in Theorem 4.1, and the proof is omitted here. A visualization of the Aitchison geometry is in Figure 4.1, which shows the latent distribution and geodesic balls on both spaces.

Theorem 4.1 (Isometry, [Egozcue et al., 2003]). *Let $\langle \cdot, \cdot \rangle_A$ denote the Aitchison inner product on $\mathring{\Delta}^D$ and $\langle \cdot, \cdot \rangle_2$ the standard inner product on $\mathbb{R}^D$. For a Helmert matrix $\mathbf{H} \in \mathbb{R}^{D \times K}$, the ILR map satisfies*

$$\langle \mathbf{x}, \mathbf{y} \rangle_A = \langle \varphi(\mathbf{x}), \varphi(\mathbf{y}) \rangle_2 \quad \forall \mathbf{x}, \mathbf{y} \in \mathring{\Delta}^D,$$

and, in particular, the ILR map is an isometry between $(\mathring{\Delta}^D, \langle \cdot, \cdot \rangle_A)$ and $(\mathbb{R}^D, \langle \cdot, \cdot \rangle_2)$.

The Riemannian metric in the tangent space of the open simplex is naturally defined by the pullback of the transformation

$$\mathbf{G}(\mathbf{x}) = \mathbf{J}_\varphi(\mathbf{x})^\top \mathbf{J}_\varphi(\mathbf{x}),$$

with $\mathbf{J}_\varphi(\mathbf{x}) = \frac{\partial \varphi(\mathbf{x})}{\partial \mathbf{x}_{1:D}}$ is computed with respect to the standard Euclidean basis. The Aitchison geodesic between $\mathbf{x}$ and $\mathbf{y}$ is computed through the pullback

$$\gamma(t) = \varphi^{-1}((1-t)\varphi(\mathbf{x}) + t\varphi(\mathbf{y})) = ((1-t) \odot \mathbf{x}) \oplus (t \odot \mathbf{y}).$$

Additional simplex-to-Euclidean bijections Aitchison and Shen [1980], Aitchison [1982] also introduced other logratio maps from the open simplex to Euclidean space. Two classical bijections are the additive logratio (ALR) and the multiplicative log ratio (MLR), defined as

$$\text{alr}(\mathbf{x})_k := \log \frac{x_k}{x_K}, \quad \text{alr}^{-1}(\mathbf{z}) = \text{softmax}([\mathbf{z}, 0]), \quad (4.2)$$

$$\text{mlr}(\mathbf{x})_k := \log \frac{x_k}{1 - \sum_{i=1}^k x_i}, \quad \text{mlr}^{-1}(\mathbf{z})_k = \frac{e^{z_k}}{\prod_{i=1}^k (1 + e^{z_i})}. \quad (4.3)$$

Note that these maps, unlike ILR, depend on the order of the input and are not isometric to Euclidean space. They produce a different map for permutations of the entries. Nevertheless, these maps are useful for reparametrization, for example the logistic-normal distribution [Aitchison and Shen, 1980] is the push-forward distribution of a Gaussian over the additive logratio map.

Stick-breaking transform A simple centering adjustment improves MLR. MLR is not centered at the vector $\frac{1}{K}\mathbf{1} \in \Delta^D$, so it does not map the zero vector in $\mathbb{R}^D$ to $\frac{1}{K}\mathbf{1}$. The stick-breaking (SB) fixes this by shifting the MLR

map:

$$\begin{aligned}\varphi : \mathring{\Delta}^D &\rightarrow \mathbb{R}^D, & \mathbf{x} &\mapsto \mathbf{z}, & z_k &= \text{mlr}(\mathbf{x})_k - \log\left(\frac{1}{K-k}\right), \\ \varphi^{-1} : \mathbb{R}^D &\rightarrow \mathring{\Delta}^D, & \mathbf{z} &\mapsto \mathbf{x}, & \mathbf{x} &= \text{mlr}^{-1}(\mathbf{y}), & y_k &= z_k + \log\left(\frac{1}{K-k}\right).\end{aligned}\tag{4.4}$$

Unlike ILR, stick-breaking remains order-dependent and is not an isometry, but it is simple, numerically stable, and widely used in probabilistic modeling [Carpenter et al., 2017].

4.1.2 Simplex stochastic interpolations

The Aitchison geometry is defined on the open simplex, but discrete observations lie on the boundary of the space. We therefore construct stochastic interpolations that move discrete labels into the interior during modeling while still allowing us to recover the original categories afterwards. Dirichlet based Gaussian process classification presented in Section 2.5.2, use a deterministic interpolation in the simplex on the parameters of the Dirichlet distribution of the form $\lambda \mathbf{e}_k + (1 - \lambda) \frac{1}{K} \mathbf{1}$, for $\lambda \in (0, 1)$.

Dirichlet interpolation We first construct stochastic interpolations where the noise process is distributed in the simplex itself. Define the distinct arg max regions

$$\mathcal{R}_k := \{\mathbf{x} \in \mathring{\Delta}^D : x_k > x_j \ \forall j \neq k\}.\tag{4.5}$$

These regions partition the interior of the simplex up to zero measure ties ($x_k = x_j$ for $j \neq k$). If the interpolated sample remains on the region $\mathcal{R}_k$ then the arg max returns the original category before interpolation. For class label $c = k$ with one-hot representation $\mathbf{e}_k$ we define the stochastic interpolation

$$\boxed{\mathbf{x} = \lambda \mathbf{e}_k + (1 - \lambda) \boldsymbol{\epsilon}}.\tag{4.6}$$

The parameter λ controls how far the sample is moved away from the boundary. Proposition 2 states that for $\lambda \geq 1/2$ the original category remains in $\mathcal{R}_k$ and is recovered via $c = \arg \max \mathbf{x}$ almost surely (See Fig. 4.2).

The noise process can be any distribution over the simplex, we choose the symmetric Dirichlet noise distribution $\text{Dir}(\alpha \mathbf{1})$ because it has a simple covariance structure in ILR coordinates. The latent ILR covariance is the same independent of the number of categories K . Let $\mathbf{z} = \varphi(\mathbf{x})$ under the ILR map, then [Aitchison, 1986]

$$\text{Cov}(\mathbf{z}) = \psi'(\alpha) \mathbf{I}_D,$$

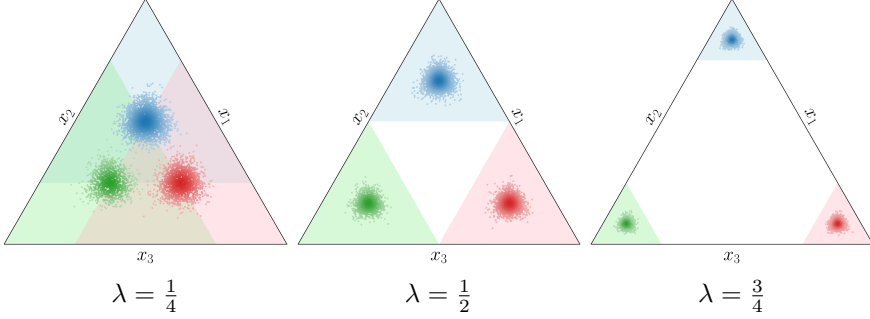


Figure 4.2: Dirichlet interpolation. The λ parameter controls the mixture distribution. For $\lambda \geq \frac{1}{2}$ the supports do not overlap and we can recover the categories almost surely. Large λ unnecessarily concentrates the mass around the simplex borders we want to avoid. $\lambda = \frac{1}{2}$ illustrates the arg max regions by the different color blocks.

where ψ' is the trigamma function. This is convenient and justifies using the same value of α for any number of categories. In practice we use $\alpha > 1$, which concentrates the interpolation's mass near the mean of the distribution and away from the boundary of the simplex.

Proposition 2 (Proposition 1, Paper III). ***Dirichlet Interpolation:** Let $\mathbf{c} = \mathbf{e}_k$ for some $k \in \{1, \dots, K\}$. Let $\mathbf{x} := \lambda \mathbf{c} + (1 - \lambda)\boldsymbol{\varepsilon}$ where $\boldsymbol{\varepsilon} \sim \text{Dir}(\boldsymbol{\alpha})$ with $\alpha_i > 0$. If $\lambda > \frac{1}{2}$, then $\arg \max \mathbf{x} = \mathbf{c}$. For $\lambda = \frac{1}{2}$, this holds almost surely under the distribution of $\boldsymbol{\varepsilon}$.*

Latent Gaussian interpolation In some cases it is more convenient to place the distribution of the noise process directly on Euclidean latent space. We do this by first interpolating in the simplex space, mapping to Euclidean space with the bijection and adding noise there. For each class k , define the interpolated point

$$\boldsymbol{\mu}^{(k)} := \lambda \mathbf{e}_k + (1 - \lambda) \frac{1}{K} \mathbf{1} \in \overset{\circ}{\Delta}^D, \quad \lambda \in (0, 1), \quad (4.7)$$

and map it to the latent mean vector $\mathbf{m}^{(k)} = \varphi(\boldsymbol{\mu}^{(k)})$. We then define the latent Gaussian interpolation

$$\boxed{\mathbf{z} \mid c = k \sim \mathcal{N}(\mathbf{m}^{(k)}, \sigma^2 \mathbf{I}_D)}. \quad (4.8)$$

The discrete labels are distributed as $c \sim \text{Cat}(\pi_1, \dots, \pi_K)$, where the class-probability vector $\boldsymbol{\pi} \in \Delta^D$ is unknown. The distribution in latent space is

a Gaussian mixture (see Fig. 4.3) centered at $\mathbf{m}^{(k)}$. This can be seen by marginalizing over the classes,

$$p(\mathbf{z}) = \sum_{k=1}^K p(c = k) p(\mathbf{z} \mid c = k) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{z} \mid \mathbf{m}^{(k)}, \sigma^2 \mathbf{I}_D).$$

The design choice is the value σ^2 . This parameter controls the overlap between the components, and we can relate this overlap to the probability of the stochastic interpolation being in the incorrect arg max region. The key insight is the isometry: distances in the latent space equal distances in the open simplex

$$\delta := d_A(\boldsymbol{\mu}^{(k)}, \boldsymbol{\mu}^{(\ell)}) = \|\mathbf{m}^{(k)} - \mathbf{m}^{(\ell)}\|_2, \quad k \neq \ell,$$

and the probability of misclassification is the probability of being closer in Euclidean distance to $\mathbf{m}^{(j)}$ than to the correct mode $\mathbf{m}^{(k)}$. Proposition 3 gives a sufficient condition on σ to make the probability of leaving the correct region (misclassification) smaller than a prescribed tolerance ε .

Proposition 3 (Proposition 1, Paper IV). *Let $K \geq 2$. Define, for each class $k \in \{1, \dots, K\}$,*

$$\mathbf{m}^{(k)} := \varphi(\boldsymbol{\mu}^{(k)}) \in \mathbb{R}^D, \quad \delta := \min_{k \neq \ell} \|\mathbf{m}^{(k)} - \mathbf{m}^{(\ell)}\|_2,$$

where δ is the Aitchison distance between the class centers on the simplex by the ILR isometry. Let $\mathcal{V}_k := \{\mathbf{z} \in \mathbb{R}^D : \|\mathbf{z} - \mathbf{m}^{(k)}\|_2 \leq \|\mathbf{z} - \mathbf{m}^{(\ell)}\|_2 \forall \ell\}$. For any $\varepsilon \in (0, 1)$, if

$$\sigma \leq \frac{\delta}{2 z_{1-\varepsilon/D}}, \quad z_q := \Phi^{-1}(q), \quad (4.9)$$

then for every $\mathbf{x} \in \mathring{\Delta}^D$ and every $k \in \{1, \dots, K\}$, the probability of miss-classification is bounded by ε ,

$$\mathbb{P}(\mathbf{Z} \notin \mathcal{V}_k \mid C = k, \mathbf{x}) \leq \varepsilon,$$

where C denotes the class label and $\mathbf{Z} \sim \mathcal{N}(\mathbf{m}^{(k)}, \sigma^2 \mathbf{I}_D)$.

4.2 Case Study III: Categorical Flow Matching

This case study combines Flow Matching introduced in Section 2.5 with the simplex-to-Euclidean bijections and the stochastic interpolations devel-

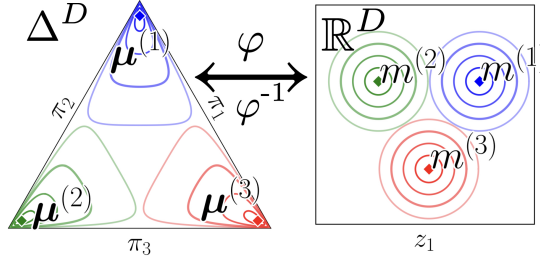


Figure 4.3: The likelihood of observing classes 1, 2, or 3 is a Gaussian mixture with modes at $\mathbf{m}^{(k)} = \varphi(\boldsymbol{\mu}^{(k)})$ in Euclidean space (right), which induces a pushforward likelihood on the simplex (left). The figure shows the likelihood for class probabilities $\boldsymbol{\pi} = (1/3, 1/3, 1/3)$.

oped in this chapter. The proposed method is Simplex-to-Euclidean Flow Matching, where the generative model is trained on Euclidean coordinates but produces samples on the simplex.

Unlike Statistical Flow Matching [Cheng et al., 2024, Davis et al., 2024], which uses the Fisher geometry of the simplex and performs Riemannian flow matching on the sphere, our method does not learn the flow on a curved manifold. Instead, we map the problem to $\mathbb{R}^D$, where Euclidean conditional flow matching is applied. This removes the need for Riemannian operations such as the exponential and logarithm maps during training and sampling.

Simplex-to-Euclidean Flow Matching (FM- Δ) Algorithm 5 details the training procedure of our method. If the data is in the interior of the simplex then the bijection is enough, but if the data is discrete on the boundary, we first use the stochastic interpolation to move the data to the interior and then apply the bijection φ and train the Euclidean flow. During sampling the arg max converts the continuous process to discrete predictions. Sampling is done with the procedure specified in Algorithm 2 with an additional arg max operator if the generated data is discrete. Any of the simplex-to-Euclidean bijections can be used within the method, and the accuracy is affected by this choice. These include additive logratio, multiplicative log ratio, stick-breaking, and isometric logratio. Note that using conditional flow matching is a design choice, the procedure is valid for any continuous Euclidean generative model, for example diffusion models [Song et al., 2021, Karras et al., 2024]. Proposition 4 justifies the correctness of FM- Δ for categorical data, it states that if the learned flow model matches

the distribution of the interpolated data with the stochastic interpolation, then the true unknown probabilities are recovered by the model's arg max estimations.

Proposition 4 (Proposition 1, Paper III). ***Categorical probabilities bound:** Let $\lambda \geq \frac{1}{2}$ so that the Dirichlet-interpolated mixture $q_\lambda(\mathbf{x}) = \sum_{k=1}^K \pi_k q_\lambda(\mathbf{x} \mid \mathbf{e}_k)$ has a.s. disjoint component supports contained in the strict arg max regions $\mathcal{R}_k := \{\mathbf{x} \in \mathring{\Delta}^D : x_k > x_j \ \forall j \neq k\}$. For any density $\tilde{q}$ on $\mathring{\Delta}^D$ define the induced (generated) categorical probabilities $\hat{\pi}_k := \int_{\mathcal{R}_k} \tilde{q}(\mathbf{x}) d\mathbf{x}$. Then the total variation between the true and generated categorical laws is bounded by the L^1 distance between the distributions:*

$$\text{TV}(\boldsymbol{\pi}, \hat{\boldsymbol{\pi}}) = \frac{1}{2} \sum_{k=1}^K |\hat{\pi}_k - \pi_k| \leq \frac{1}{2} \|\tilde{q} - q_\lambda\|_1.$$

In particular, $\|\tilde{q} - q_\lambda\|_1 \rightarrow 0$ implies $\hat{\boldsymbol{\pi}} \rightarrow \boldsymbol{\pi}$ in total variation, and the arg max discretization of $\tilde{q}$ recovers exactly the true categorical distribution.

Algorithm 5 Training of Simplex-to-Euclidean Flow Matching

Require: Data $\mathbf{c} \in \Delta^D$, weight $\lambda \in (0, 1)$, Dirichlet $\alpha > 0$, bijection φ , base p_0 (e.g. $\mathcal{N}(\mathbf{0}, \mathbf{I})$), coupling π (indep. or minibatch OT), isDiscrete

```

1: for each mini-batch do
2:   for each  $\mathbf{c}$  in batch do
3:     if isDiscrete then
4:       Sample  $\boldsymbol{\varepsilon} \sim \text{Dir}(\alpha, \dots, \alpha)$ 
5:        $\mathbf{x} \leftarrow \lambda \mathbf{c} + (1 - \lambda) \boldsymbol{\varepsilon}$ 
6:     else
7:        $\mathbf{x} \leftarrow \mathbf{c}$ 
8:        $\mathbf{z}_1 \leftarrow \varphi(\mathbf{x})$  ▷ To Euclidean space
9:       Sample  $\mathbf{z}_0 \sim p_0$ ; pair  $(\mathbf{z}_0, \mathbf{z}_1)$  via  $\pi$ 
10:      Sample  $t \sim \text{Unif}(0, 1)$ 
11:       $\mathbf{z}_t \leftarrow (1 - t)\mathbf{z}_0 + t\mathbf{z}_1, \quad \mathbf{u}_t \leftarrow \mathbf{z}_1 - \mathbf{z}_0$ 
12:      Update  $\theta$  by  $\min \|\mathbf{v}_\theta(\mathbf{z}_t, t) - \mathbf{u}_t\|^2$  (Eq. (2.18))
```

4.2.1 Empirical evaluation

We highlight two experiments from Paper III that illustrate the two parts of the construction. The first considers a density supported on the open simplex, where the role of the bijection and geometry can be seen directly. The second is a language-generation task on discrete data, where the full pipeline including stochastic interpolation is needed. The main comparison methods are LinearFM which uses Euclidean geometry on the simplex itself and Statistical Flow Matching (SFM) [Cheng et al., 2024, Davis et al., 2024].

Density on the open simplex This experiment considers checkerboard data projected to the open simplex. The data is continuous in Δ^2 , so no interpolation is needed in this comparison. The experiment studies the effect of the geometric representation (simplex, simplex-to-sphere, or simplex-to-Euclidean) and two different choices of bijection (SB or ILR). Figure 4.4 shows that our method produces samples that align more closely with the target density than both LinearFM and SFM. In particular, the Euclidean baseline on the simplex and the Fisher-sphere construction produce many more invalid samples near the corners, whereas the simplex-to-Euclidean approach better preserves the structure of the target. Both the isometric logratio and stick-breaking bijections work well, with stick-breaking performing slightly better in this example.

Language generation We consider the Text8 dataset [Mahoney, 2011], which consists of $K = 27$ categories: 26 letters and the space character, split into random chunks of size 256. This task illustrates the full discrete-data setting, and we rely on Dirichlet interpolation to move the data to the open simplex, followed by a Euclidean flow model in latent coordinates. We use an auto-regressive model to evaluate the NLL and compute the entropy [Wang and Komatsuzaki, 2021]; the NLL should be low and the entropy over generated data should be close to the entropy of the test data. Table 4.1 shows the NLL, entropy, and difference in entropy. It shows that our method is the most competitive among the continuous approaches and improves over LinearFM and SFM. However, the best overall result is obtained by the discrete-state diffusion model SEDD [Lou et al., 2024]. SEDD is a discrete diffusion model that operates directly in the discrete space. At the time of submission of our method there was still a gap between latent continuous models and discrete diffusion models, which has been closing recently [Lee et al., 2026]. The outcome is that Simplex-to-Euclidean Flow Matching is a strong continuous baseline, even if it does not fully close the gap to the best discrete-state models.

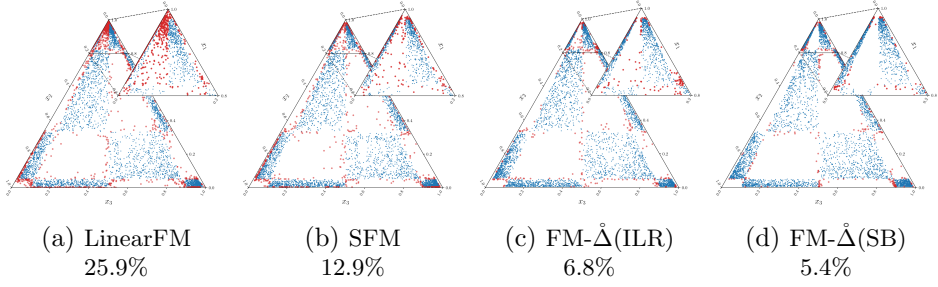


Figure 4.4: Samples from Checkerboard on the simplex. Red points indicate samples not aligned with the true density (% indicated in caption). The zoomed area shows the top region $x_1 \geq \frac{4}{5}$, emphasizing the differences.

4.3 Case Study IV: Gaussian Process Classification

This case study uses the isometric logratio and the latent Gaussian interpolation to build a multiclass Gaussian process classifier that is conjugate in latent space. Similar to Case Study III, the idea is to map the simplex-valued problem to Euclidean coordinates and then apply a standard continuous model there, in this case the model is Gaussian process regression. We present the method and show with empirical studies that it is accurate and well-calibrated.

Isometric Logratio Gaussian Process Let the training data be $\{(\mathbf{r}_i, c_i)\}_{i=1}^N$, where $\mathbf{r}_i$ denotes the regressor and $c_i \in \{1, \dots, K\}$ the class label. The construction uses the ingredients introduced earlier in the chapter. Each label $c_i = k$ is moved to the interior of the simplex with the interpolation

$$\boldsymbol{\mu}^{(k)} = \lambda \mathbf{e}_k + (1 - \lambda) \frac{1}{K} \mathbf{1},$$

and then mapped to Euclidean space with the ILR bijection (Eq. 4.1) φ giving the latent $\mathbf{m}^{(k)} = \varphi(\boldsymbol{\mu}^{(k)})$. The latent Gaussian likelihood (Eq. 4.8) together with the misclassification probability bound (Prop. 3), turns classification into GP regression in $\mathbb{R}^D$. The model is:

$$\boxed{\mathbf{f}(\cdot) \sim \mathcal{GP}(\mathbf{0}, K_{\theta}), \quad \mathbf{z}_i \mid \mathbf{f}(\mathbf{r}_i) \sim \mathcal{N}(\mathbf{f}(\mathbf{r}_i), \sigma^2 \mathbf{I}_D).} \quad (4.10)$$

Since the prior and the likelihood are Gaussian, the model is conjugate in $\mathbb{R}^D$ and posterior inference is exact in $\mathbb{R}^D$. The model uses D latent processes because of the dimensionality of the space. Compared to Dirichlet-based GP classification (Section 2.5) it requires one less latent process and

Table 4.1: Text8: NLL and Entropy on the GPT-J-6B model. The baseline results are taken from the respective papers.

	Model	NLL ↓	Entropy	Diff
Discrete	MDLM	6.76	7.55	+0.07
	DFM	6.78	7.58	+0.10
	D3PM	6.93	7.38	-0.10
	SEDD	6.49	7.17	-0.31
	MultiFlow	6.73	7.39	-0.09
Continuous	LinearFM	7.35	7.62	+0.14
	α -Flow($\alpha=-0.5$)	7.14	7.37	-0.11
	α -Flow($\alpha=0.5$)	7.00	7.42	-0.06
	α -Flow($\alpha=1$)	7.09	7.51	+0.03
	SFM	6.85	7.38	-0.10
	FM- $\hat{\Delta}$ (ILR)	6.81	7.39	-0.09
	FM- $\hat{\Delta}$ (ILR) _{w/OT}	6.89	7.38	-0.10
	Data	4.10	7.48	0

removes the need for a distributional approximation in the model construction.

For a new regressor $\mathbf{r}_*$, class-probability predictions are obtained from the expected value after mapping back to the simplex with φ^{-1} . Predictions are estimated with Monte Carlo

$$\mathbb{E}[\boldsymbol{\pi}_* \mid \mathbf{r}_*, \mathcal{D}] \approx \frac{1}{S} \sum_{s=1}^S \varphi^{-1}(\mathbf{f}_*^{(s)}) \quad (4.11)$$

$$\mathbf{f}_*^{(s)} \sim \mathcal{N}(\boldsymbol{\mu}_*, \boldsymbol{\Sigma}_*).$$

The method’s cubic complexity in the size of the training data limits its scalability. For larger data sizes, Paper IV studies two inducing-point sparse variants. *Uncollapsed-ILR* is a sparse variational classifier that parametrizes the probabilities as $\boldsymbol{\pi} = \varphi^{-1}(\mathbf{z})$; the objective factorizes over observations, which allows for mini-batch optimization [Hensman et al., 2015]. *Collapsed-ILR* instead keeps the Gaussian process regression in latent space, which is specified with the Collapsed GP regression approximation (see Eq. 2.24). The first variant models the probabilities directly while the second approximates the latent GP regression.

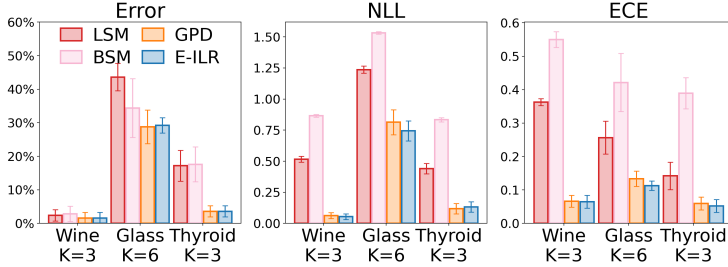


Figure 4.5: Error, NLL and ECE for UCI datasets in the exact setting.

4.3.1 Empirical evaluation

We highlight two experimental evaluations from Paper IV. The objective of the experiments is to benchmark our work in terms of accuracy and calibration with respect to the existing literature. The first studies exact latent inference on small UCI classification tasks, where the full GP posterior can be computed exactly. The second studies larger UCI classification tasks for which we use inducing-point methods for inference. The most important related comparison method is Dirichlet-based Gaussian process (GPD). We additionally include comparisons against models that achieve posterior conjugacy through data augmentation; these are Logistic softmax (LSM) [Galy-Fajou et al., 2020] and Bijective softmax (BSM) [Galy-Fajou, 2022]. We denote by E-ILR our model with exact latent inference, by C-ILR the sparse Collapsed version, and by U-ILR the sparse Uncollapsed version of our method. In all experiments we report classification error, negative log-likelihood (NLL), and expected calibration error (ECE).

Exact UCI regression We first evaluate the exact ILR model on three small UCI classification data sets, where cubic-time GP inference is still feasible. Figure 4.5 shows that our method and Dirichlet-based GP classification achieve similar error rates, but the ILR model attains lower NLL on two of the three tasks and the lowest ECE on all three. This means that the exact ILR construction is at least as accurate as the main conjugate baseline, while giving slightly more calibrated predictive probabilities. The auxiliary-variable methods are consistently worse in this setting.

Sparse UCI regression We next consider seven larger UCI classification benchmarks, where inducing-point approximations are required. Figure 4.6 summarizes the results. Overall, Collapsed-ILR and sparse Dirichlet GP classification are the strongest methods across the three metrics. The two

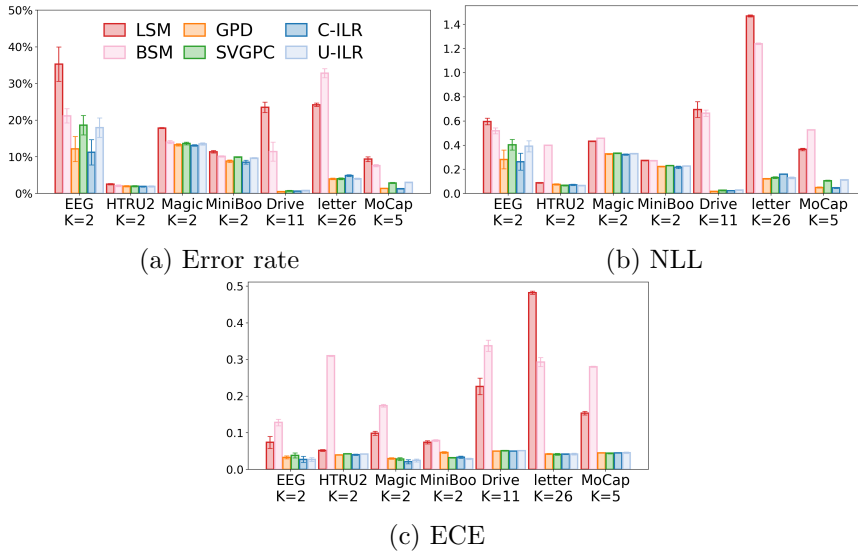


Figure 4.6: Sparse models performance on UCI datasets.

ILR-based sparse models are competitive with sparse GPD and are often slightly better calibrated. In particular, the ILR-based sparse methods obtain lower ECE on several tasks, and on the LETTER data set Uncollapsed-ILR performs better than Collapsed-ILR. The data-augmented softmax baselines struggle here, especially as the number of classes increases. The main conclusion is that the sparse ILR constructions preserve the good calibration of the exact model while scaling to larger problems.

Chapter 5

Conclusion

Summary of findings The findings in this thesis tie to the research questions posed in the introduction. Approximate Bayesian inference can be improved by geometry-aware modeling if the metric is chosen *carefully*. We considered metrics constructed from the target distribution. Paper I concluded that the Riemannian Laplace approximation with the Fisher information matches strongly curved targets most accurately in small-dimensional problems. Paper II found that the correct choice of metric inside geodesic slice sampling captures curvature. If the target is multimodal, sampling using the inverse metrics improves movement between the modes.

Simplex geometry can be connected to Euclidean representations through stochastic interpolations and simplex-to-Euclidean bijections. Paper III found that this makes Euclidean generative models applicable to discrete data, while allowing exact recovery of the original category with the arg max operation. Paper IV showed that the Aitchison geometry together with the latent stochastic interpolation converts the problem into Gaussian process regression, which enables exact latent inference and gives accurate and calibrated predictions.

Overall, the thesis incorporated the geometry of the statistical problem into the method itself, which led to improved accuracy (while making it clear when these gains come at a computational cost) or to a latent reinterpretation of the original problem. Next we give limitations of the methods presented and avenues for future work.

Limitations and Outlook The main limitation of Papers I and II is computational complexity: numerical integration is expensive, and each integration step becomes cubic in complexity for the Fisher information

metric. Future work can study (i) approximations of the geodesics, for example Taylor approximations [Hartmann et al., 2023], and (ii) amortization of the geodesics, for example geometry-aware variational approximations [Chevallier et al., 2022].

A limitation of Paper II is the lack of a guarantee of mode coverage. A more detailed theoretical study could show under which conditions the sampler can generate geodesics that reach other modes.

The main limitation of Paper III is that the method does not scale as well as discrete diffusion with respect to K . Recent work has shown that continuous diffusion does scale as well as discrete diffusion with additional improvements [Roos et al., 2026, Lee et al., 2026]. Future work can explore whether training the flow directly with the negative categorical log likelihood (cross entropy) and time reparameterization can improve our method in the same way. Another distinction with discrete diffusion is that some of these methods model the score only for a subset of categories [Lou et al., 2024], whereas we model all categories at the same time with the velocity field. It might be that restricting the velocity field to move between a subset of categories or to lower-dimensional projections improves the scalability of our method [Chen et al., 2026].

Paper IV shares the limitations of Gaussian process regression: it scales purely with the training data even with sparse approximations. Since our method is compatible with advances in Gaussian process regression, future work can test the scalability of the method on large classification tasks under scalable Gaussian process regression methods [Maroñas and Hernández-Lobato, 2023].

The simplex-to-Euclidean bijections and the Aitchison geometry remain largely underutilized in modern machine learning. The sampling of densities constrained by a linear system. Diederer and Zamboni [2025] construct a general bijection from convex polytopes to Euclidean space (which uses the ILR map). This can be used as a reparametrization of the target distribution defined on a convex set. In a similar way, high-dimensional categorical distributions can be converted to Euclidean target distributions with our procedure, which can ease sampling [Mohanty and Zhang, 2026].

Classification with the softmax function leads to neural collapse [Kothapalli, 2023]. Future work can assess whether the ILR map avoids this behavior. This would open up the possibility of computing geodesics in the ILR-latent space, which might lead to improvements in the study of latent geometric representations [Yu et al., 2025].

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